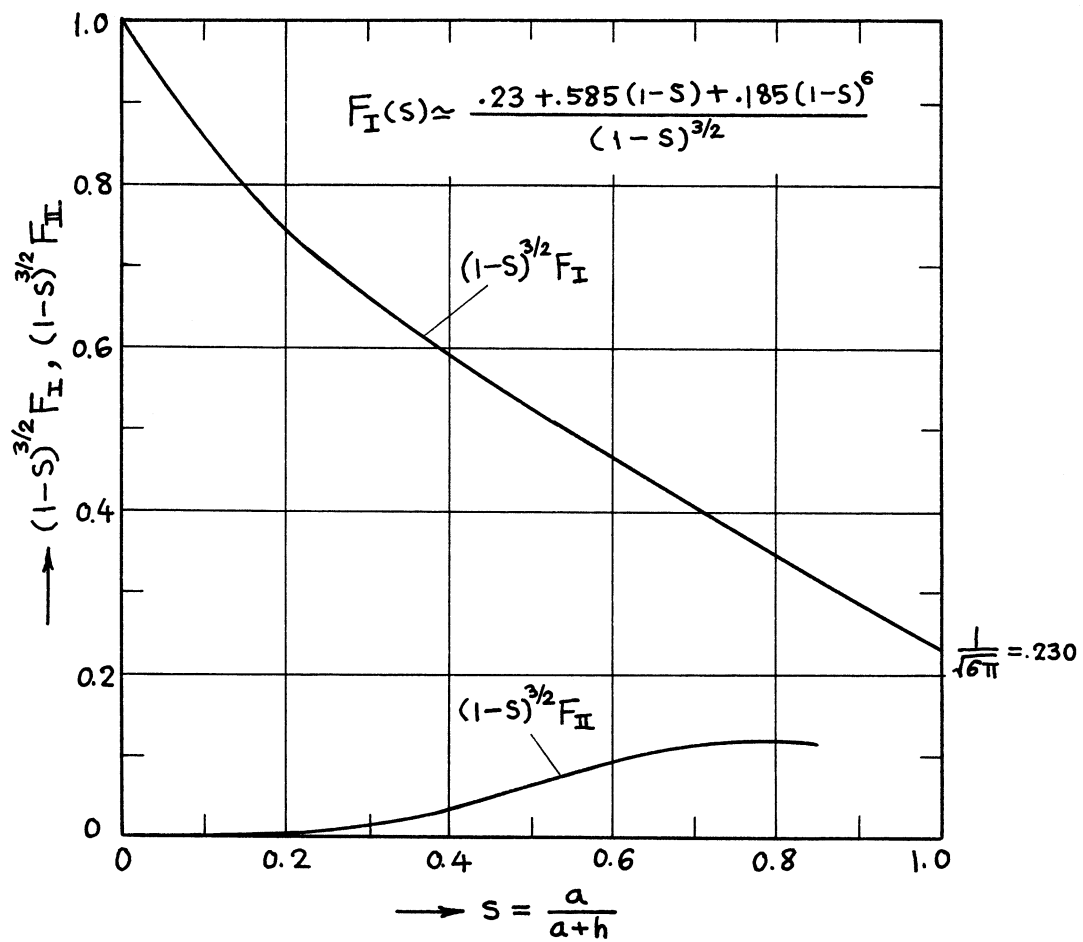
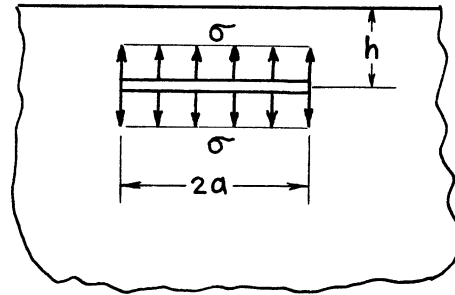


$$K_I = \sigma\sqrt{\pi a} \cdot F_I(s)$$

$$K_{II} = \sigma\sqrt{\pi a} \cdot F_{II}(s)$$

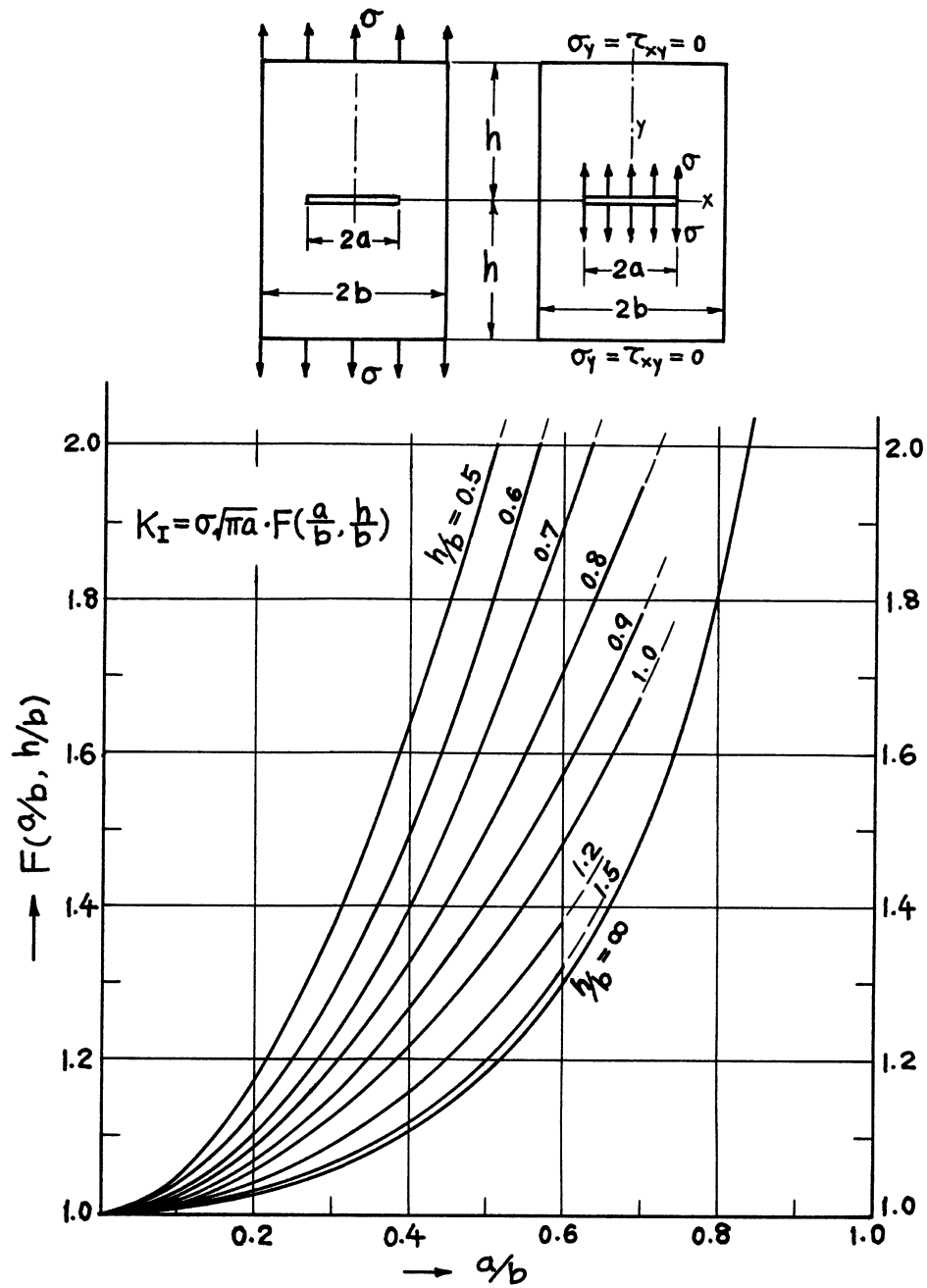
$$s = \frac{a}{a+h}$$



Methods: Singular Integral Equations, Beam Theory ($s \rightarrow 1$)

Accuracy: Better than 1%

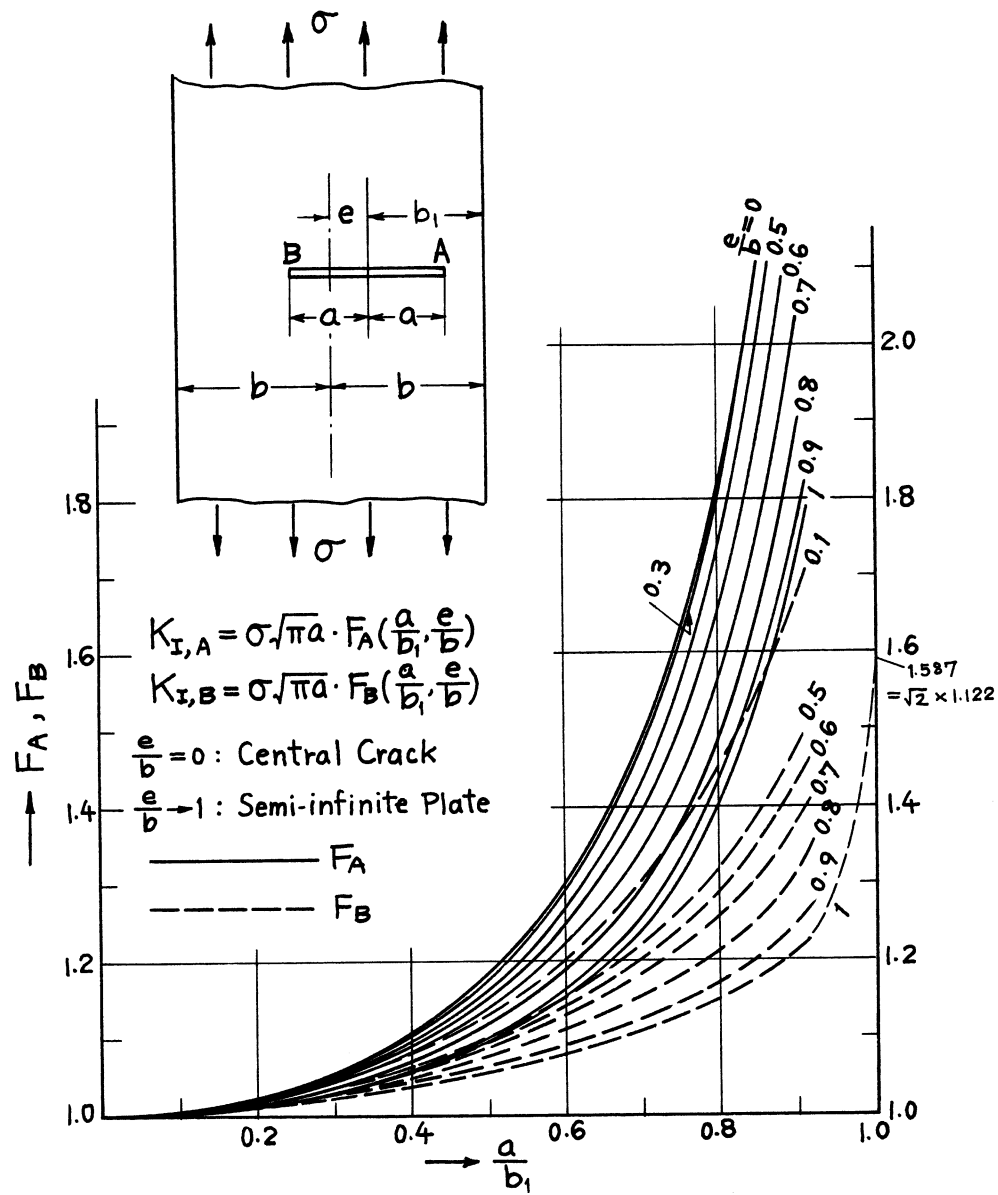
References: **Erdogan 1973, Tada 1985**



Methods: Expansions of Complex Stress Potentials Combined with a Boundary Collocation Method (Isida), Boundary Collocation Method (Kobayashi)

Accuracy: Better than 1%

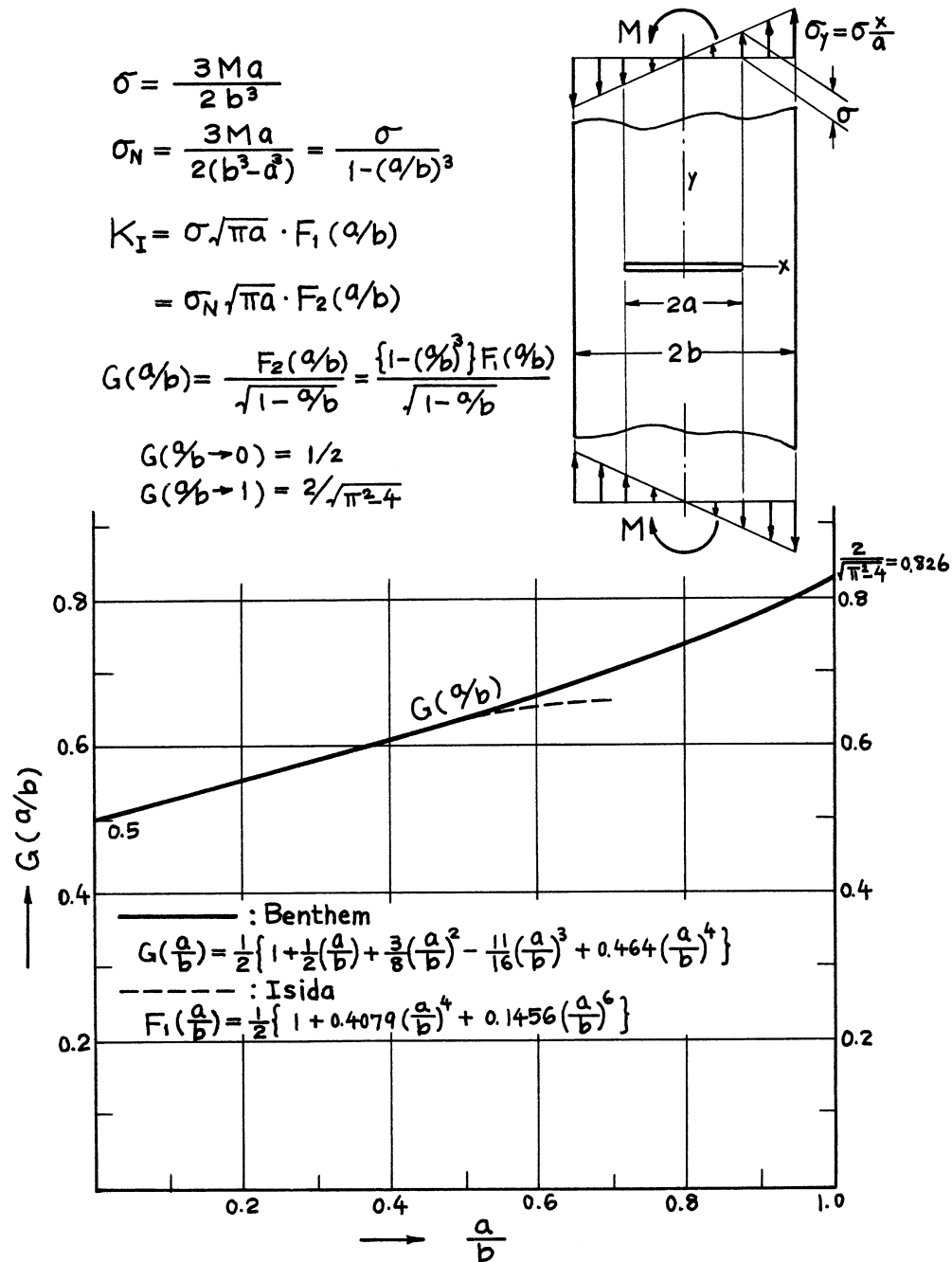
References: **Kobayashi 1964; Isida 1971a,b**



Method: Laurent's Expansions of Complex Stress Potentials

Accuracy: Better than 1%

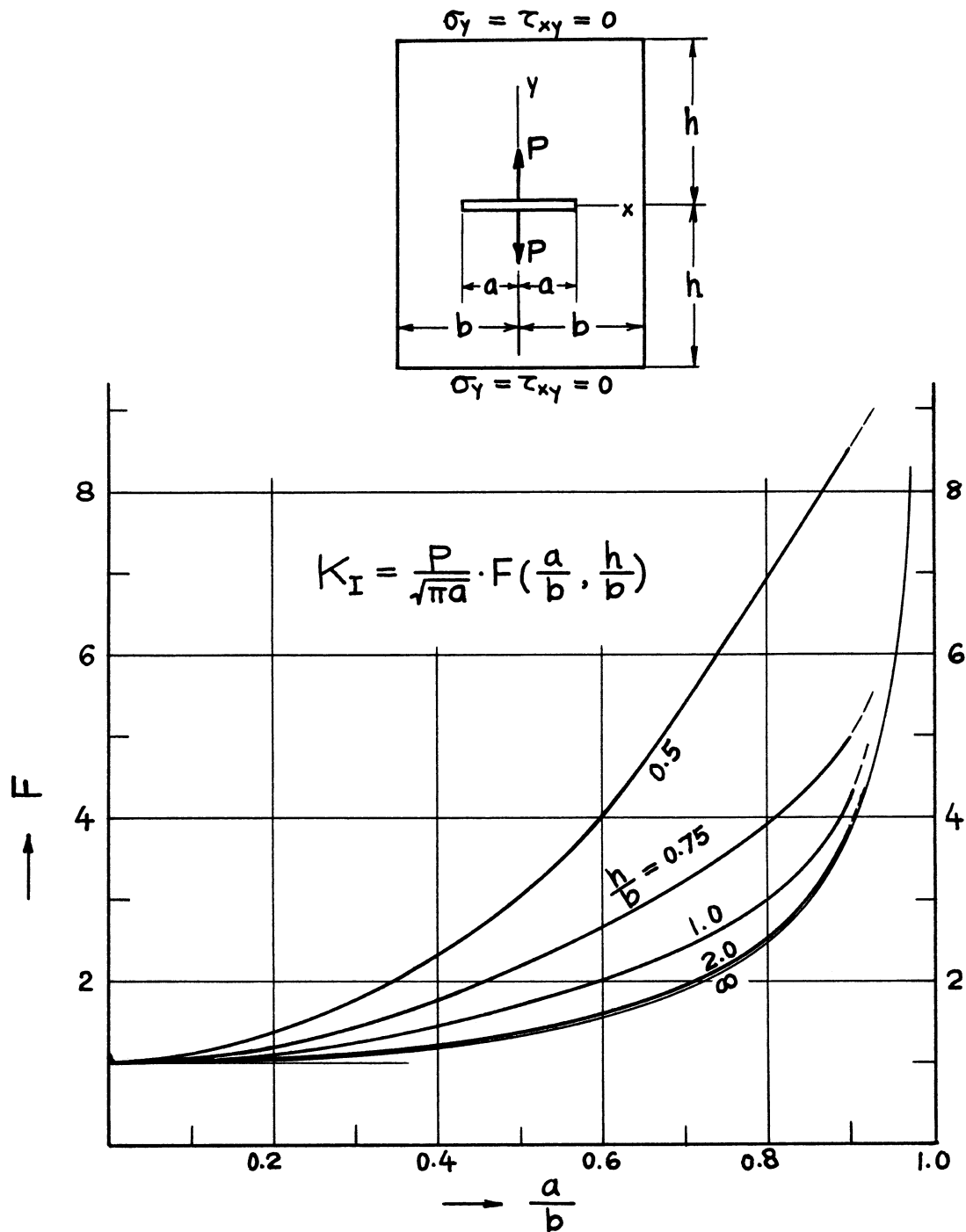
Reference: **Isida 1965a**



Method: From Stress Concentration Factor for an Elliptical Hole (Isida), Asymptotic Approximation (Benthem)

Accuracy: Better than 1% [$G(a/b)$ formula by Benthem]

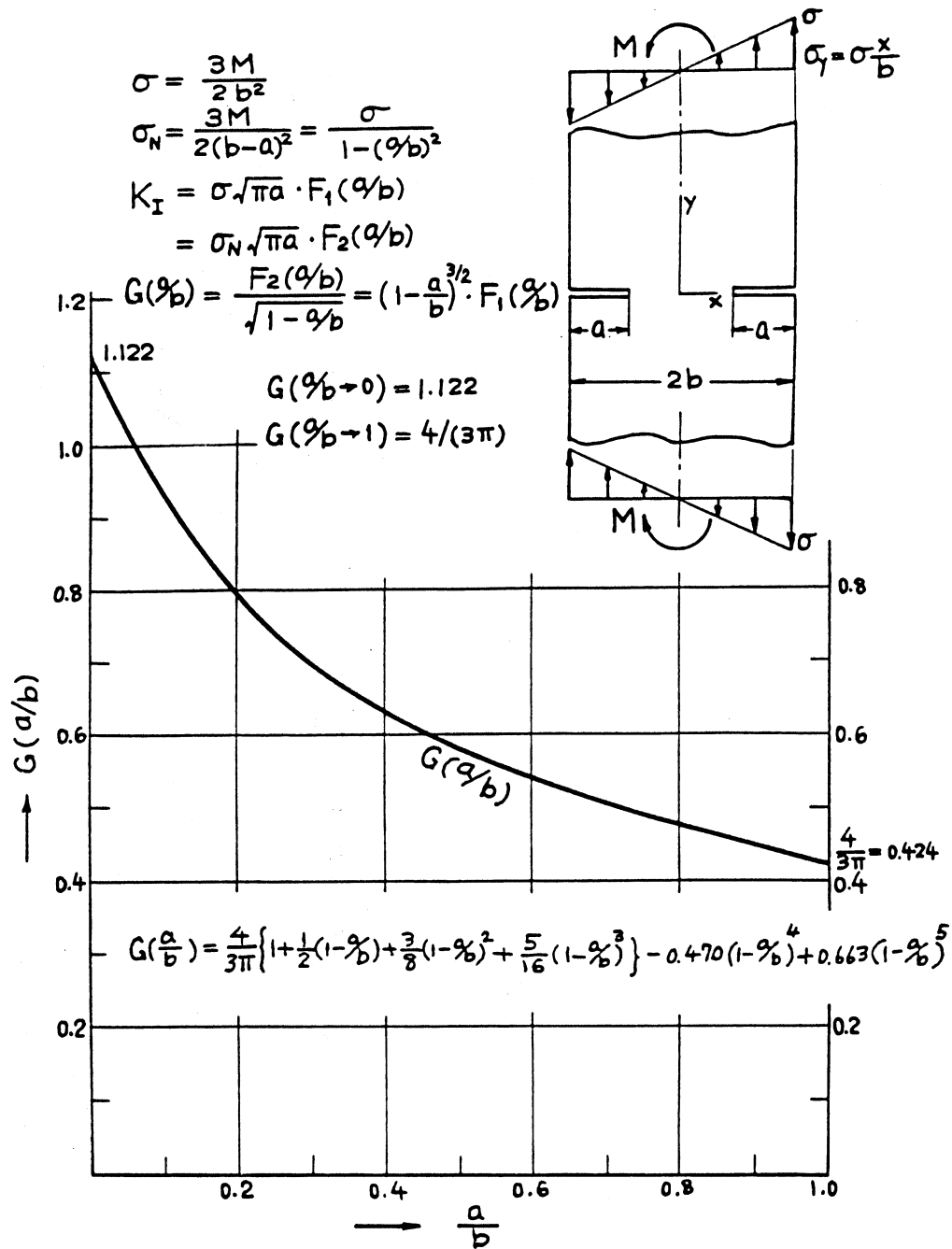
References: **Isida 1956; Benthem 1972**



Method: Boundary Collocation Method

Accuracy: Curves were drawn based on results having better than 0.1% accuracy.

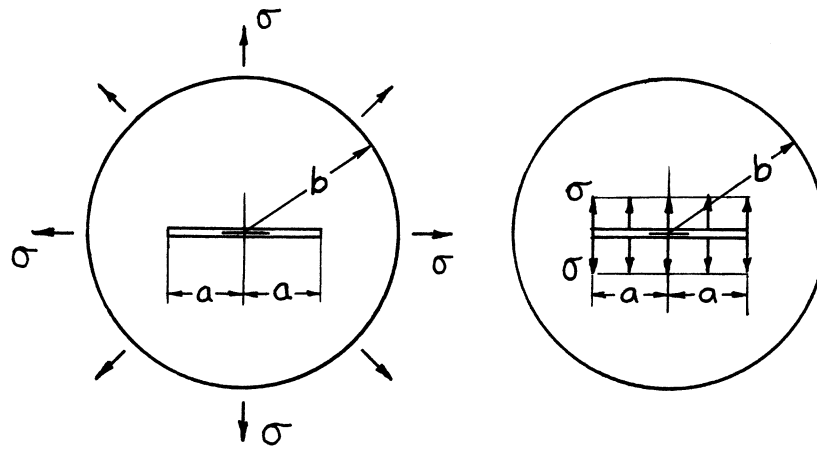
Reference: **Newman 1971**



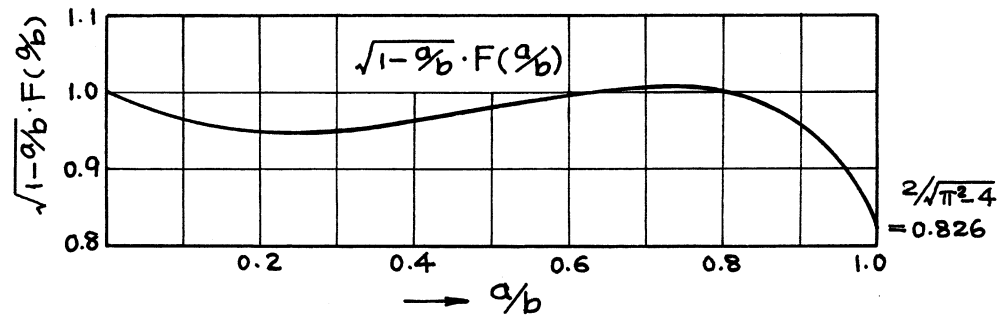
Method: Asymptotic Approximation

Accuracy: Better than 1%

Reference: **Benthem 1972**



$$K_I = \sigma \sqrt{\pi a} \cdot F(a/b)$$

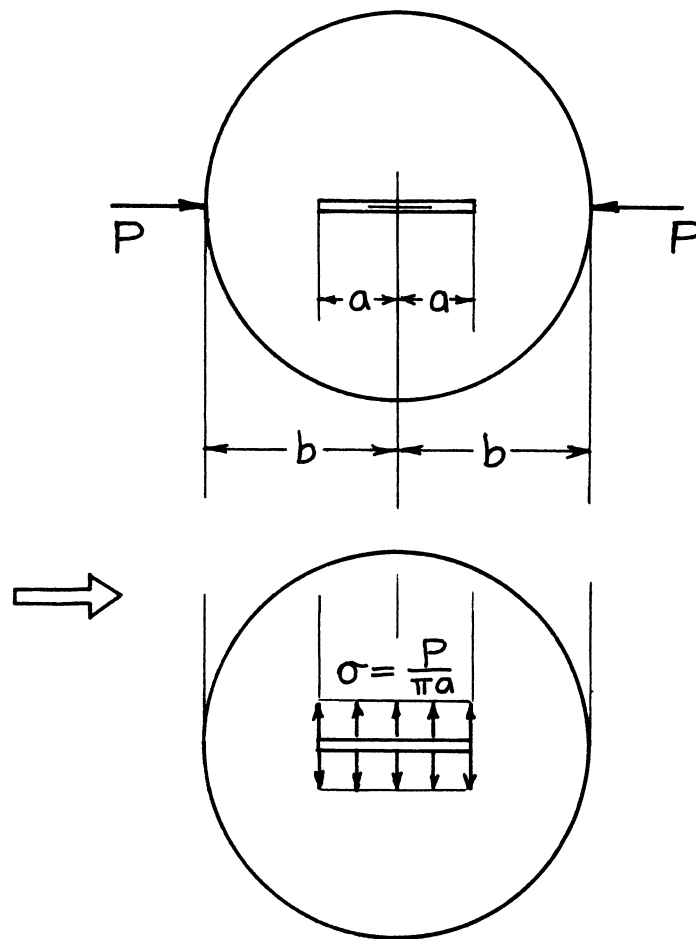


Methods: Mapping Collocation Technique (Bowie), Integral Transform (Tweed)

Accuracy: Curve was drawn based on the values having four significant figures for $0 \leq a/b \leq 0.9$.

References: **Bowie 1970b; Tweed 1972a**

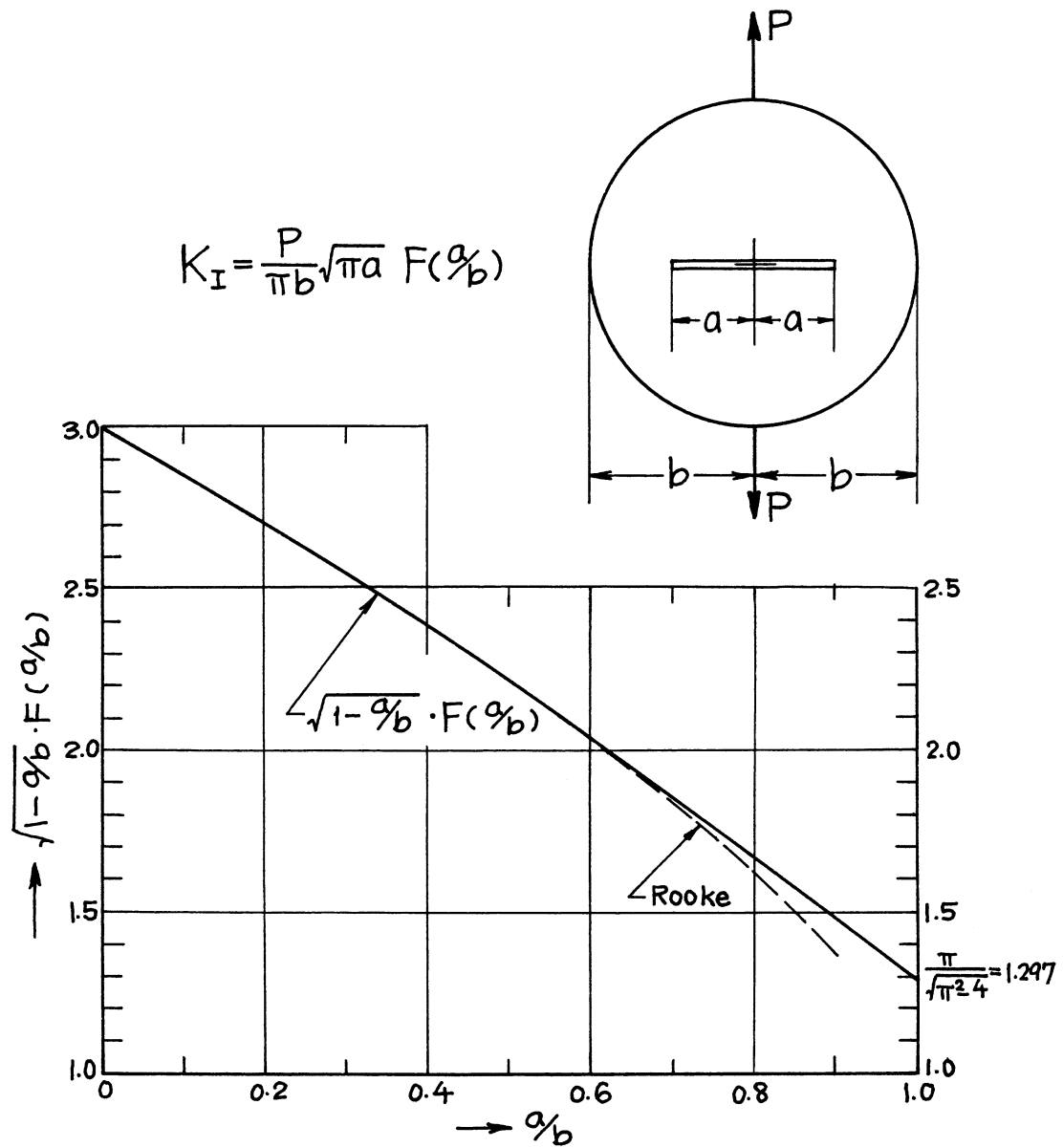
NOTE: For eccentrically located radial internal cracks, see **Tweed 1972a**.



$$K_I = \frac{P}{\pi b} \sqrt{\pi a} \cdot F\left(\frac{a}{b}\right)$$

(See page 11.6 for $F(a/b)$.)

Additional References to page 11.6: Libatskii 1965, 1967; Yamera 1965. For eccentrically located radial cracks, see Tweed 1972a.

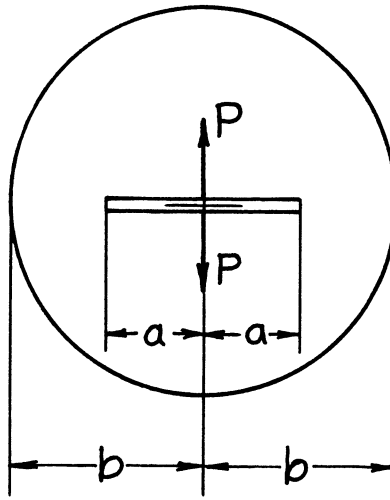


Methods: Mellin Transform and Integral Equation (Rooke, $a/b \leq 0.6$), Interpolated Asymptotically (Tada, $a/b > 0.6$)

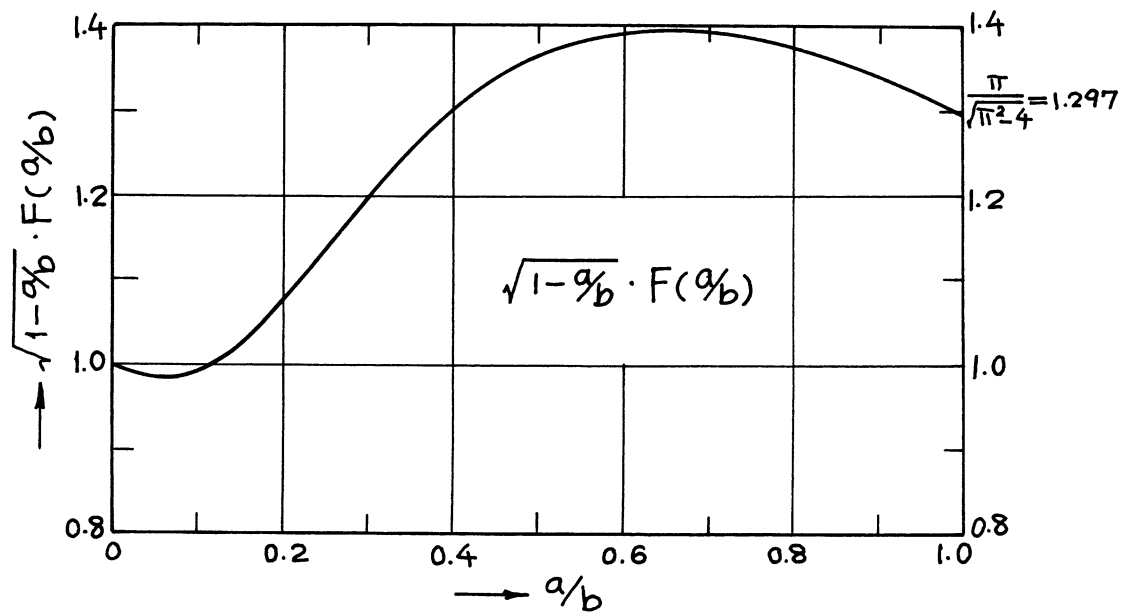
Accuracy: Better than 1%

References: **Rooke 1973b; Tada 1973**

NOTE: For eccentrically located radial cracks, see **Rooke 1973b**.



$$K_I = \frac{P}{\sqrt{\pi a}} \cdot F(a/b)$$



Method: Estimated Asymptotically
 Accuracy: Expected to be better than 2%
 Reference: **Tada 1973**

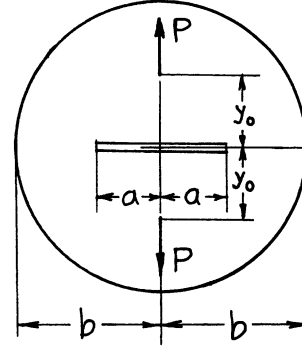
$$K_I = \frac{P}{\sqrt{\pi a}} \cdot F(a/b, y_0/b, \nu), \quad 0 < y_0/b < 1$$

$$\left(K_I = \frac{P}{\sqrt{\pi a}} \cdot F(a/b, y_0/b = 0, 1) \right)$$

$$a/b \rightarrow 0: F \rightarrow \frac{a}{b} \left\{ (1 + \alpha) \left(\frac{b}{y_0} + \frac{y_0}{b} \right) - (2\alpha - 1) \left(\frac{y_0}{b} \right)^3 \right\}, \quad 0 < \frac{y_0}{b} \leq 1$$

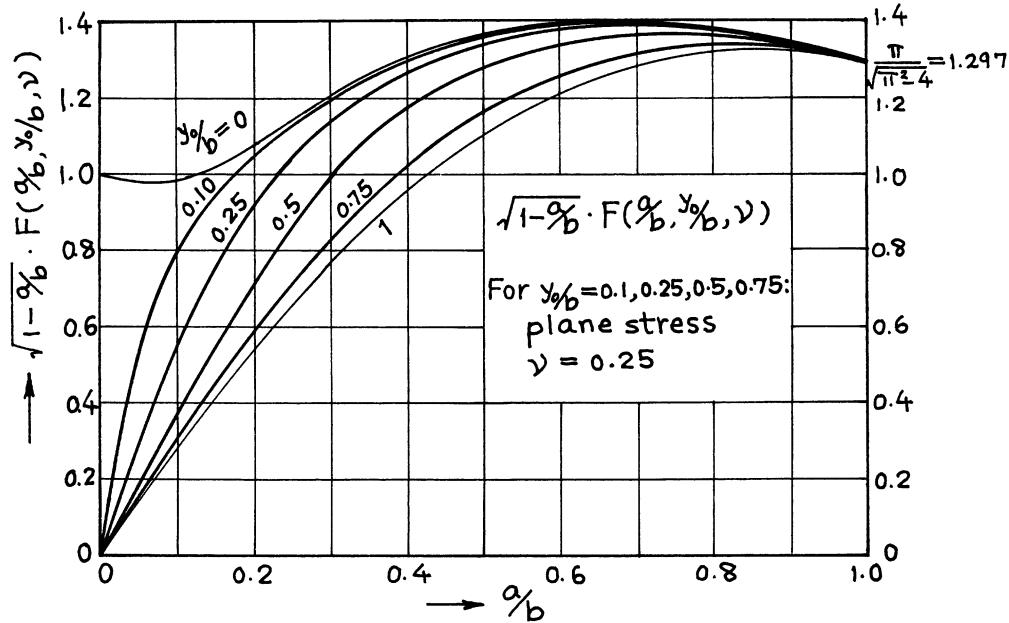
$$F \rightarrow 1, y_0/b = 0.$$

$$a/b \rightarrow 1: F \rightarrow \frac{\pi}{\sqrt{\pi^2 - 4}} \cdot \frac{1}{\sqrt{1 - a/b}}, \quad 0 \leq \frac{y_0}{b} \leq 1$$



where

$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

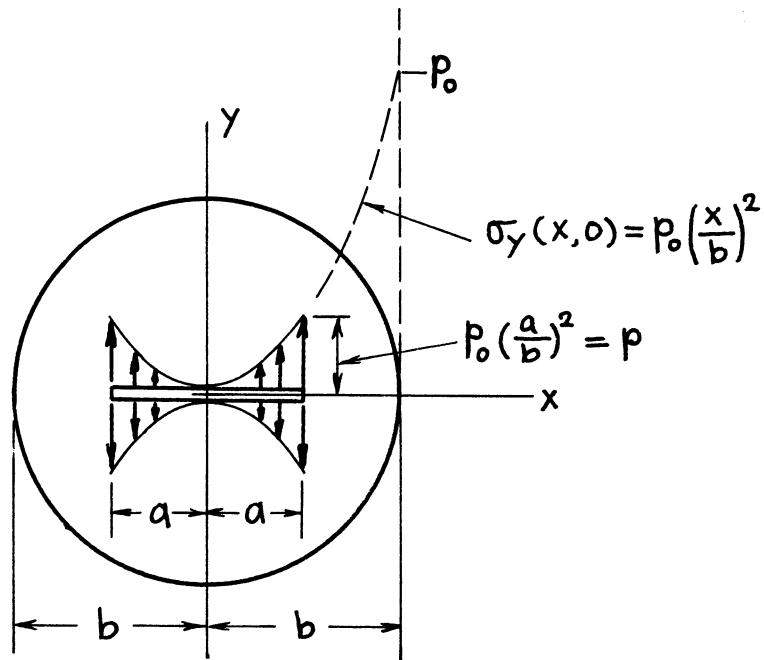


Methods: Integral Equation (**Likatskii**, $y_0/b = 0.25, 0.5, 0.75$, $a/b < 0.7$), Interpolated Asymptotically (Tada, $a/b \geq 0.7$ and $y_0/b = 0, 1$)

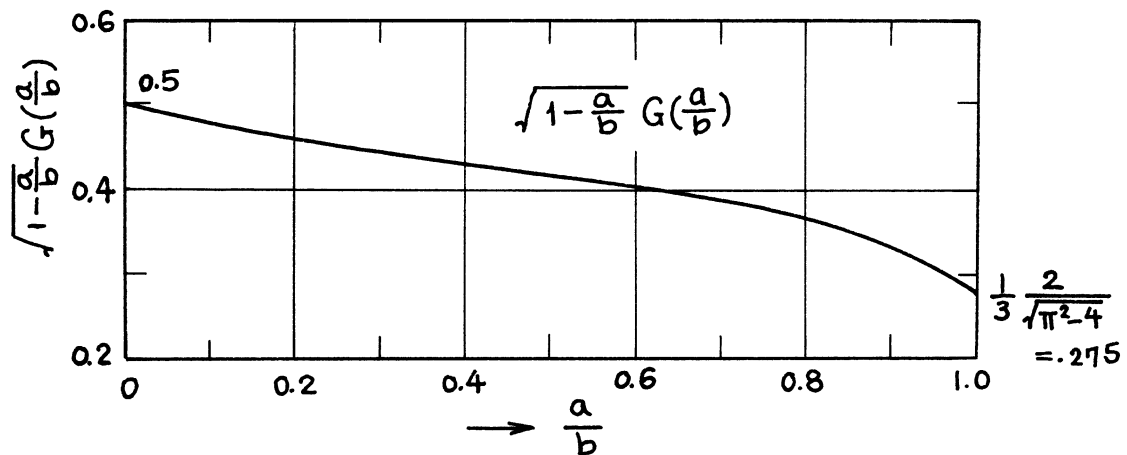
Accuracy: Better than 2% for any a/b and y_0/b

References: **Libatskii 1967; Tada 1973**

NOTE: For special cases $y_0/b = 0$ and 1, see **pages 11.9** and **page 11.8**, respectively.



$$K_I = P\sqrt{\pi a} \cdot G\left(\frac{a}{b}\right)$$

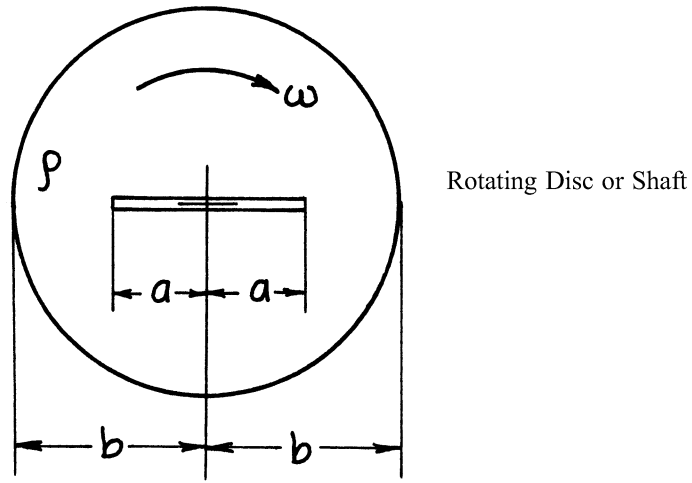


Method: Reduced from the results for **page 11.12** (Rooke: Integral Transform, Plane Strain, $\nu = 0.2, 0.3, 0.4$; Isida: Series Expansion, Plane Stress, $\nu = 0.3$)

Accuracy: 1%

References: **Rooke 1972; Isida 1974; Tada 1985**

NOTE: For an application, see **page 11.12**.



$$K_I = \frac{\alpha + 1}{4} \rho \omega^2 b^2 \sqrt{\pi a} \cdot \left\{ F(a/b) - \frac{3\alpha - 1}{\alpha + 1} \left(\frac{a}{b} \right)^2 G(a/b) \right\}$$

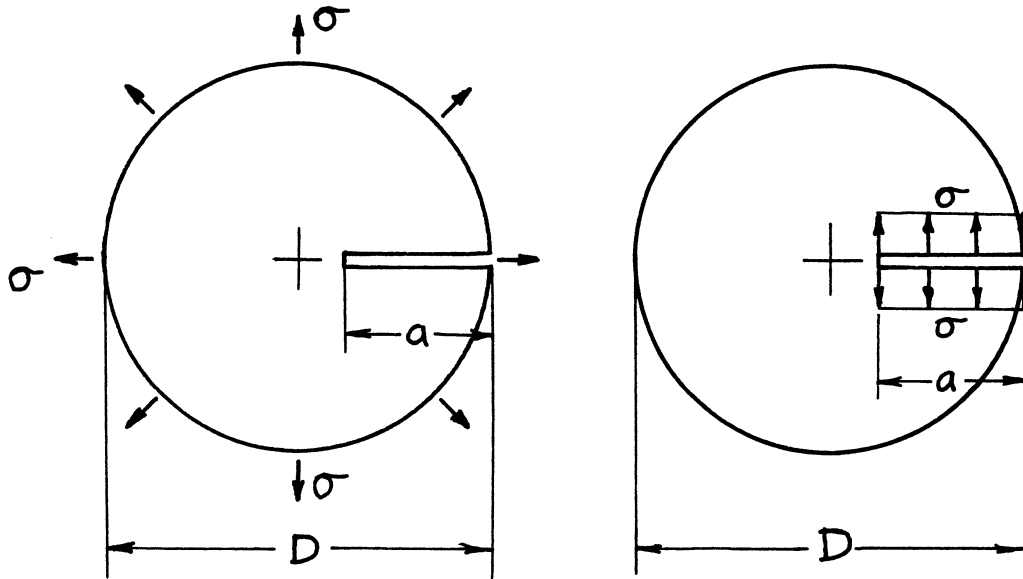
where

ω = angular velocity

ρ = density

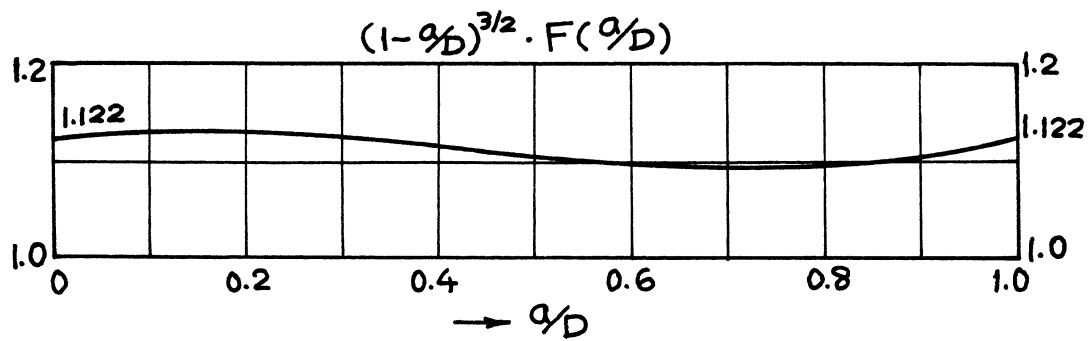
$$\alpha = \begin{cases} \frac{1}{2}(1 + \nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1 - \nu}\right) & \text{plane strain} \end{cases}$$

For numerical values of $F(a/b)$ and $G(a/b)$, and more information, see **pages 11.6 and 11.11**. For eccentrically located radial cracks, see **Rooke 1972**.



$$K_I = \sigma \sqrt{\pi a} \cdot F(a/D)$$

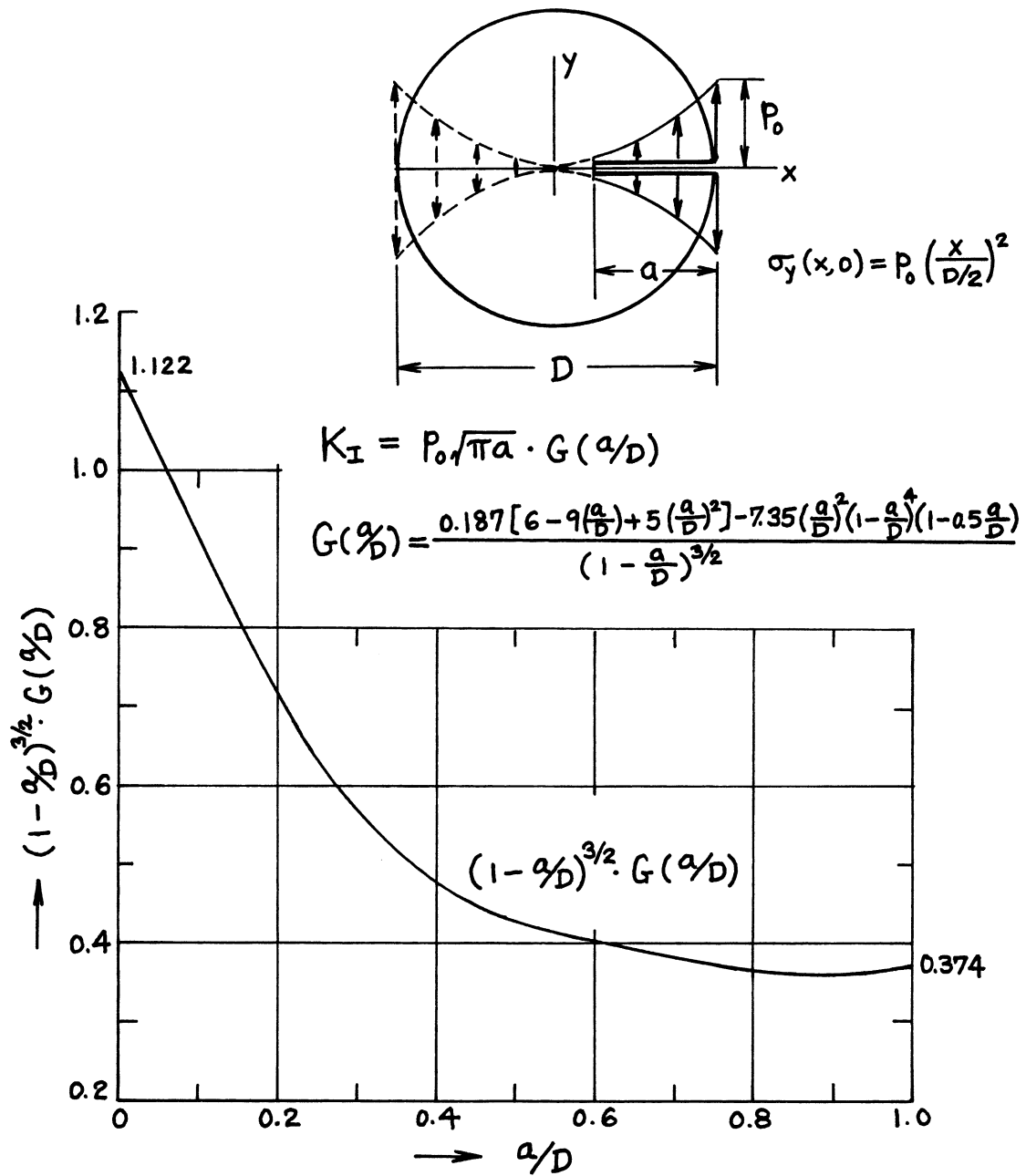
$$F(a/D) = \frac{1.122 + 0.140(a/D) - 0.545(a/D)^2 + 0.405(a/D)^3}{(1 - a/D)^{3/2}}$$



Methods: Mellin Transform and Integral Equation (Tweed, $a/D \leq 0.5$), Asymptotic Interpolation (**Tada**)

Accuracy: Better than 1%

References: **Tweed 1973; Tada 1973**

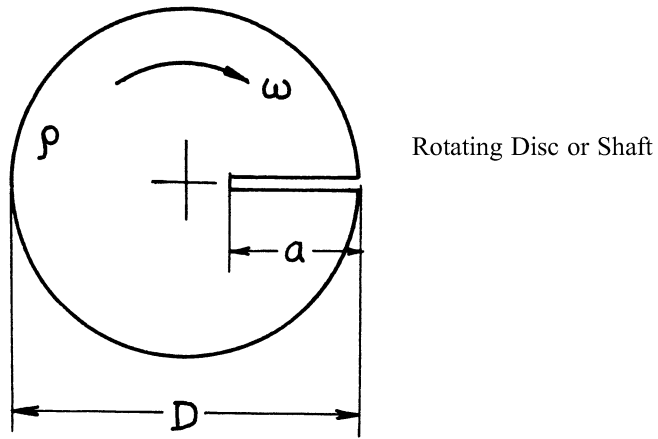


Methods: Extracted from the results of **page 11.15** (Rooke; Integral Transform; $a/D \leq 0.5$),
Influence Function (Sire)

Accuracy: 1%

References: **Rooke 1973a; Sire 1989; Tada 2000**

NOTE: For an application, see **page 11.15**.



$$K_I = \frac{1+\alpha}{16} \rho \omega^2 D^2 \sqrt{\pi a} \left\{ F(a/D) - \frac{3\alpha-1}{1+\alpha} G(a/D) \right\}$$

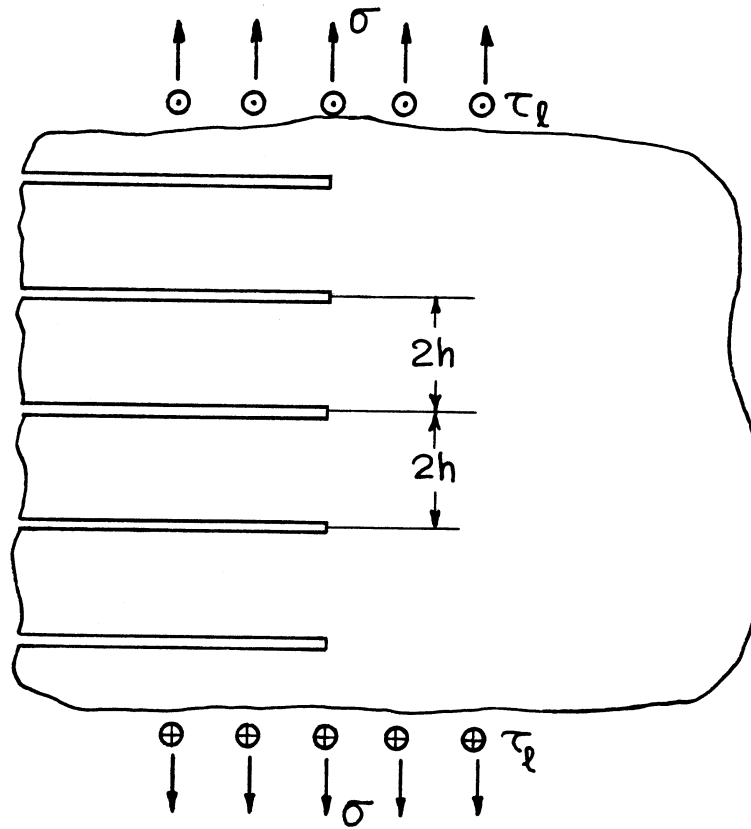
where

ω = angular velocity

ρ = density

$$\alpha = \begin{cases} \frac{1}{2}(1+\nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

For numerical values of $F(a/D)$ and $G(a/D)$ and more information, see **pages 11.13 and 11.14**.



$$K_I = \sigma\sqrt{h}$$

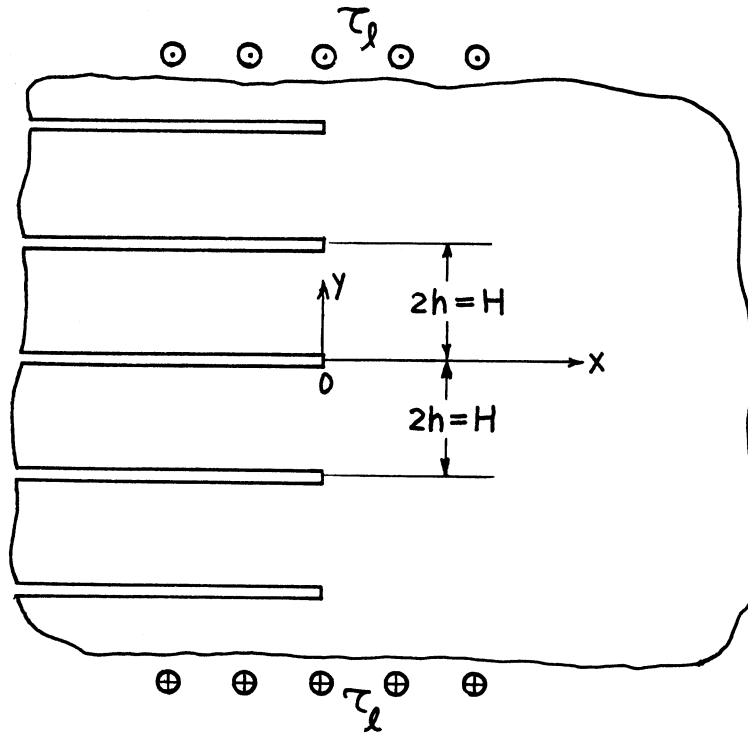
$$K_{III} = \tau_l\sqrt{2h}$$

Methods: K_I Fourier Transform (Koiter, Benthem), Energy Consideration (Paris, Rice); K_{III} Westergaard Stress Function (Tada)

Accuracy: Exact

References: **Koiter 1956, 1961; Paris 1955, 1960; Rice 1967; Benthem 1972; Tada 1973**

NOTE: For complete Mode III stress functions, see **page 12.2**.



$$z = x + iy$$

$$Z_{III}(z) = \frac{\tau_\ell}{\sqrt{1 - e^{-\frac{2\pi z}{H}}}}$$

$$\bar{Z}_{III}(z) = \tau_\ell \cdot \frac{H}{\pi} \cosh^{-1} \left(e^{\frac{\pi \bar{z}}{H}} \right)$$

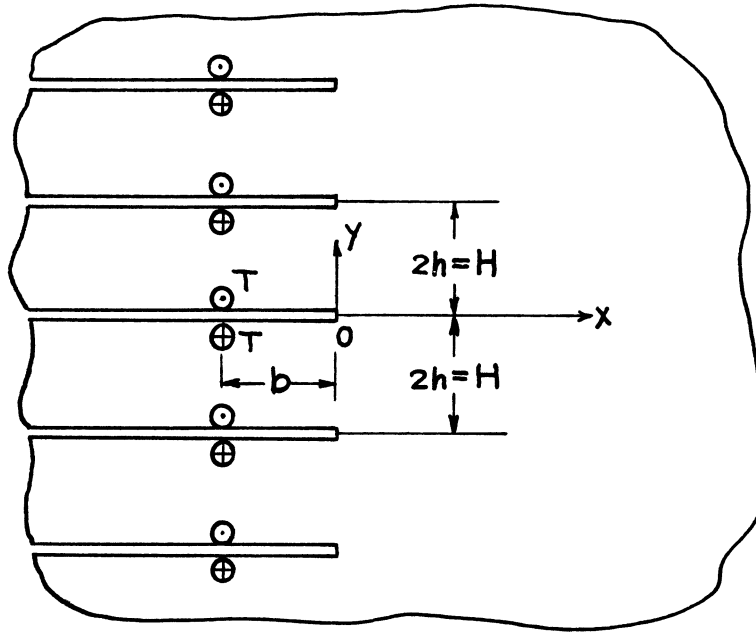
$$K_{III} = \tau_\ell \sqrt{H}$$

$$w(x, 0) = \frac{\tau_\ell}{G} \cdot \frac{H}{\pi} \cos^{-1} \left(e^{\frac{\pi x}{H}} \right)_{x \leq 0}$$

Method: Westergaard Stress Function

Accuracy: Exact

Reference: **Tada 1973**



$$z = x + iy$$

$$Z_{III}(z) = \frac{2T}{H} \cdot \frac{e^{-\frac{\pi b}{H}} \sqrt{1 - e^{-\frac{2\pi b}{H}}}}{\left(e^{\frac{2\pi x}{H}} - e^{-\frac{2\pi b}{H}} \right) \sqrt{1 - e^{-\frac{2\pi x}{H}}}}$$

$$\bar{Z}_{III}(z) = \frac{2T}{\pi} \tan^{-1} \sqrt{\frac{e^{\frac{2\pi b}{H}} - 1}{1 - e^{-\frac{2\pi x}{H}}}}$$

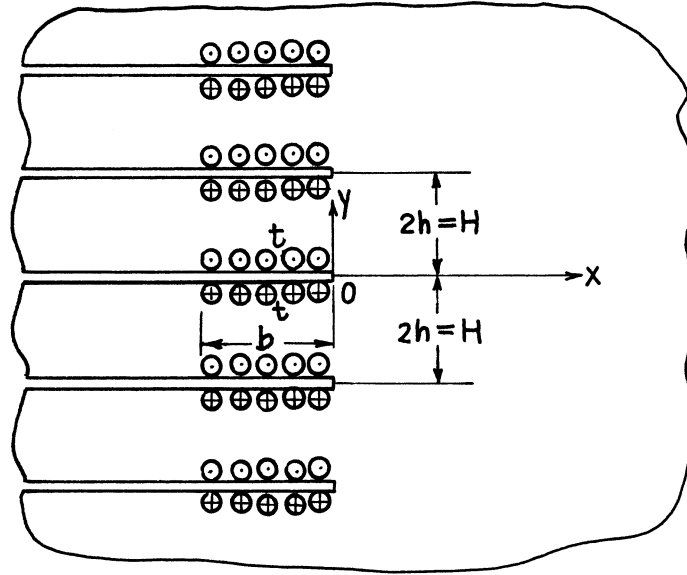
$$K_{III} = \frac{2T}{\sqrt{H}} \frac{1}{\sqrt{e^{\frac{2\pi b}{H}} - 1}}$$

$$w(x, 0) = \frac{2T}{\pi G} \left(\tanh^{-1} \right) \sqrt{\frac{1 - e^{\frac{2\pi b}{H}}}{1 - e^{-\frac{2\pi x}{H}}}} \left(\begin{array}{l} x < -b \\ -b < x < 0 \end{array} \right)$$

Method: Westergaard Stress Function

Accuracy: Exact

Reference: **Tada 1973**



$$z = x + iy$$

$$Z_{III}(z) = \frac{2t}{\pi} \left\{ \frac{\cos^{-1} \left(e^{-\frac{\pi b}{H}} \right)}{\sqrt{1 - e^{-\frac{2\pi x}{H}}}} - \tan^{-1} \sqrt{\frac{e^{\frac{2\pi b}{H}} - 1}{1 - e^{-\frac{2\pi x}{H}}}} \right\}$$

$$\bar{Z}_{III}(z) = \frac{2t}{\pi} \int_0^b \tan^{-1} \sqrt{\frac{e^{\frac{2\pi x_0}{H}} - 1}{1 - e^{-\frac{2\pi x}{H}}}} dx_0$$

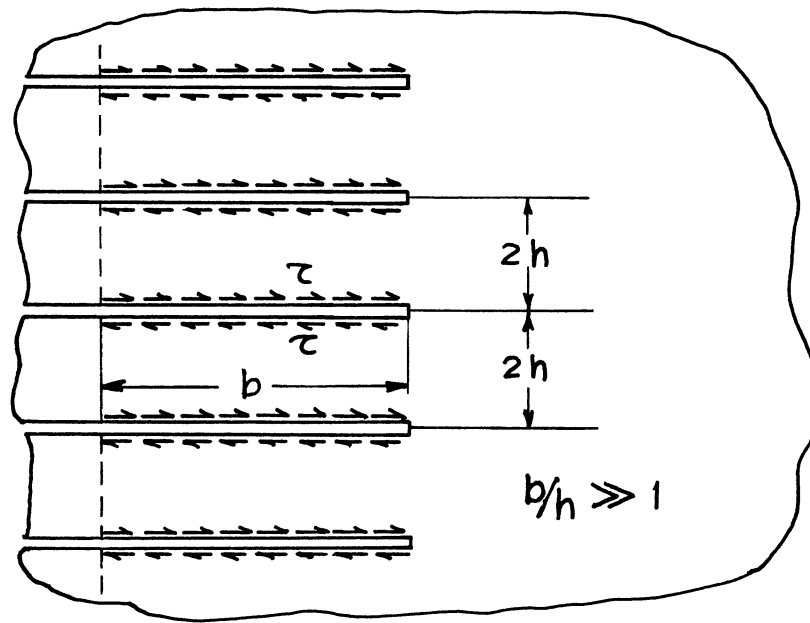
$$K_{III} = \frac{2t}{\pi} \sqrt{H} \cos^{-1} \left(e^{-\frac{\pi b}{H}} \right)$$

$$w(x, 0) = \frac{t}{\pi G} \int_0^b \ln \left| \frac{\sqrt{1 - e^{\frac{2\pi x_0}{H}}} + \sqrt{1 - e^{-\frac{2\pi x}{H}}}}{\sqrt{1 - e^{\frac{2\pi x_0}{H}}} - \sqrt{1 - e^{-\frac{2\pi x}{H}}}} \right| dx_0$$

Method: Westergaard Stress Function

Accuracy: Exact

Reference: **Tada 1973**



$$K_{II} = \tau \sqrt{3h} \left(\frac{b}{h} + 0.2865 \right)$$

Method: Fourier Transform

Accuracy: Numerical value 0.2865 is believed to have accuracy within one unit of the last digit (obtained by numerical integration).

Reference: **Benthem 1972**

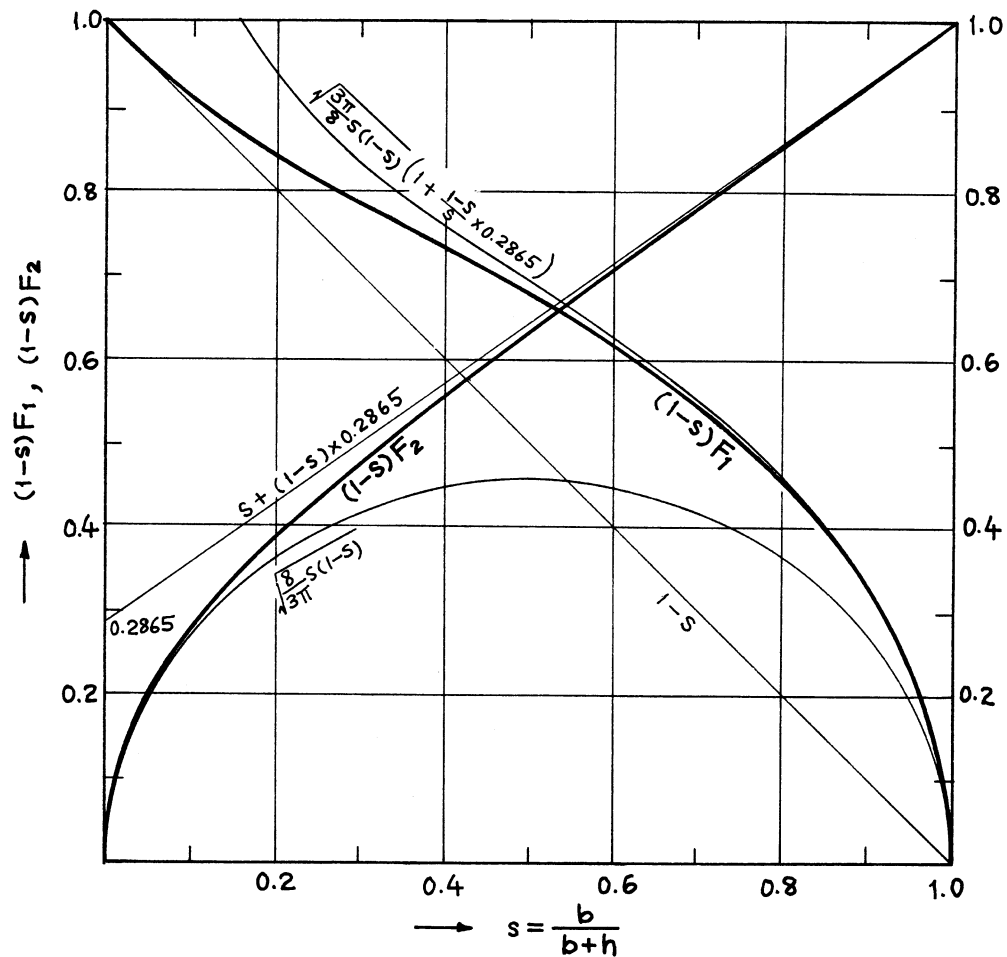
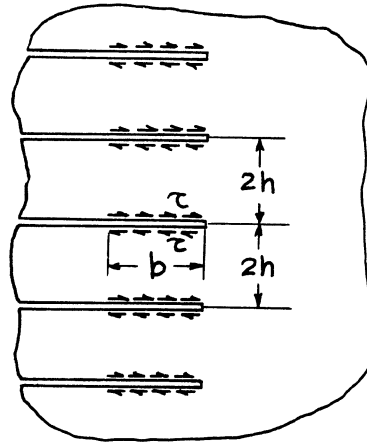
$$s = \frac{b}{b+h} = \frac{b/h}{1+b/h} = \frac{1}{1+h/b}$$

$$K_{II} = \frac{2}{\pi} \tau \sqrt{2\pi b} \cdot F_1(s) \\ = \tau \sqrt{3h} \cdot F_2(s)$$

$$G(s) = \sqrt{1-s} F_1(s) = \frac{1-s}{\sqrt{\frac{8s}{3\pi}}} F_2(s)$$

$$s \rightarrow 0 \left(\frac{b}{h} \rightarrow 0 \right): G = 1$$

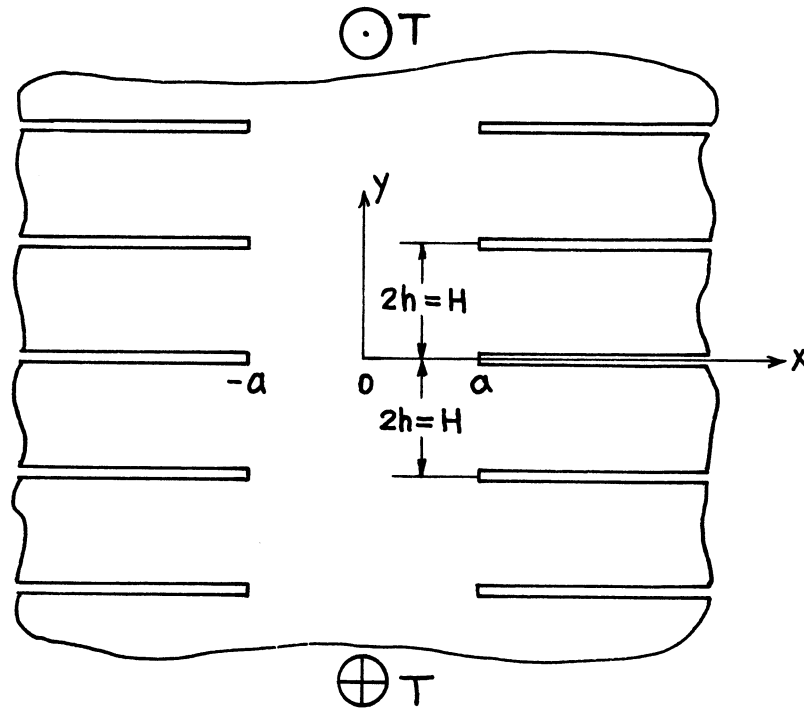
$$s \rightarrow 1 \left(\frac{b}{h} \rightarrow \infty \right): \sqrt{\frac{3\pi}{8s}} \{ s + (1-s) \times 0.2865 \}$$



Method: Estimated Asymptotically

Accuracy: Expected to be better than 3%

Reference: Tada 1973



$$z = x + iy$$

$$Z_{III}(z) = \frac{2T}{H} \frac{1}{\sqrt{\left(\cosh \frac{\pi a}{H} / \cosh \frac{\pi z}{H}\right)^2 - 1}}$$

$$\bar{Z}_{III}(z) = \frac{2T}{\pi} \sin^{-1} \left(\frac{\sinh \frac{\pi z}{H}}{\sinh \frac{\pi a}{H}} \right)$$

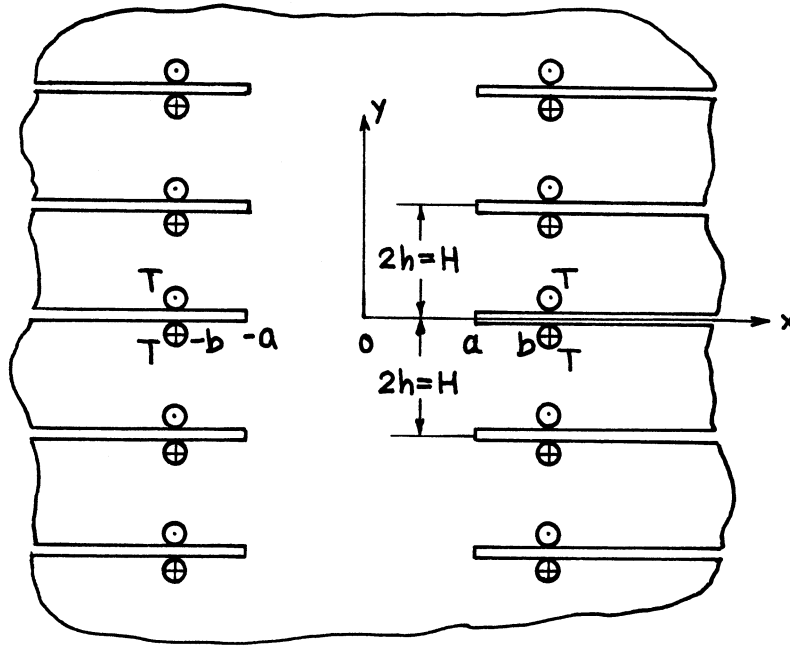
$$K_{III} = \frac{2T}{H} \sqrt{H \coth \frac{\pi a}{H}}$$

$$w(x, 0) = \frac{2T}{\pi G} \sinh^{-1} \left(\frac{\sinh \frac{\pi x}{H}}{\sinh \frac{\pi a}{H}} \right) \quad |x| \geq a$$

Method: Westergaard Stress Function

Accuracy: Exact

Reference: **Tada 1973**



$$z = x + iy$$

$$Z_{III}(z) = \frac{2T}{H} \frac{\sinh \frac{\pi b}{H} \sqrt{\left(\cosh \frac{\pi b}{H}\right)^2 - \left(\cosh \frac{\pi a}{H}\right)^2}}{\left\{ \left(\cosh \frac{\pi b}{H}\right)^2 - \left(\cosh \frac{\pi z}{H}\right)^2 \right\} \sqrt{\left(\cosh \frac{\pi a}{H} / \cosh \frac{\pi z}{H}\right)^2 - 1}}$$

$$\bar{Z}_{III}(z) = \frac{2T}{\pi} \tan^{-1} \frac{\sqrt{1 - \left(\sinh \frac{\pi a}{H} / \sinh \frac{\pi b}{H}\right)^2}}{\left(\sinh \frac{\pi a}{H} / \sinh \frac{\pi z}{H}\right)^2 - 1}$$

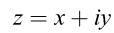
$$K_{III} = \frac{2T}{H} \sqrt{H \coth \frac{\pi a}{H}} \frac{\sinh \frac{\pi b}{H}}{\sqrt{\left(\cosh \frac{\pi b}{H}\right)^2 - \left(\cosh \frac{\pi a}{H}\right)^2}}$$

$$w(x, 0) = \frac{2T}{\pi G} \left(\frac{\tanh^{-1}}{\coth^{-1}} \right) \sqrt{\frac{1 - \left(\sinh \frac{\pi a}{H} / \sinh \frac{\pi b}{H}\right)^2}{1 - \left(\sinh \frac{\pi a}{H} / \sinh \frac{\pi x}{H}\right)^2}} \begin{pmatrix} |x| > b \\ a \leq |x| < b \end{pmatrix}$$

Method: Westergaard Stress Function

Accuracy: Exact

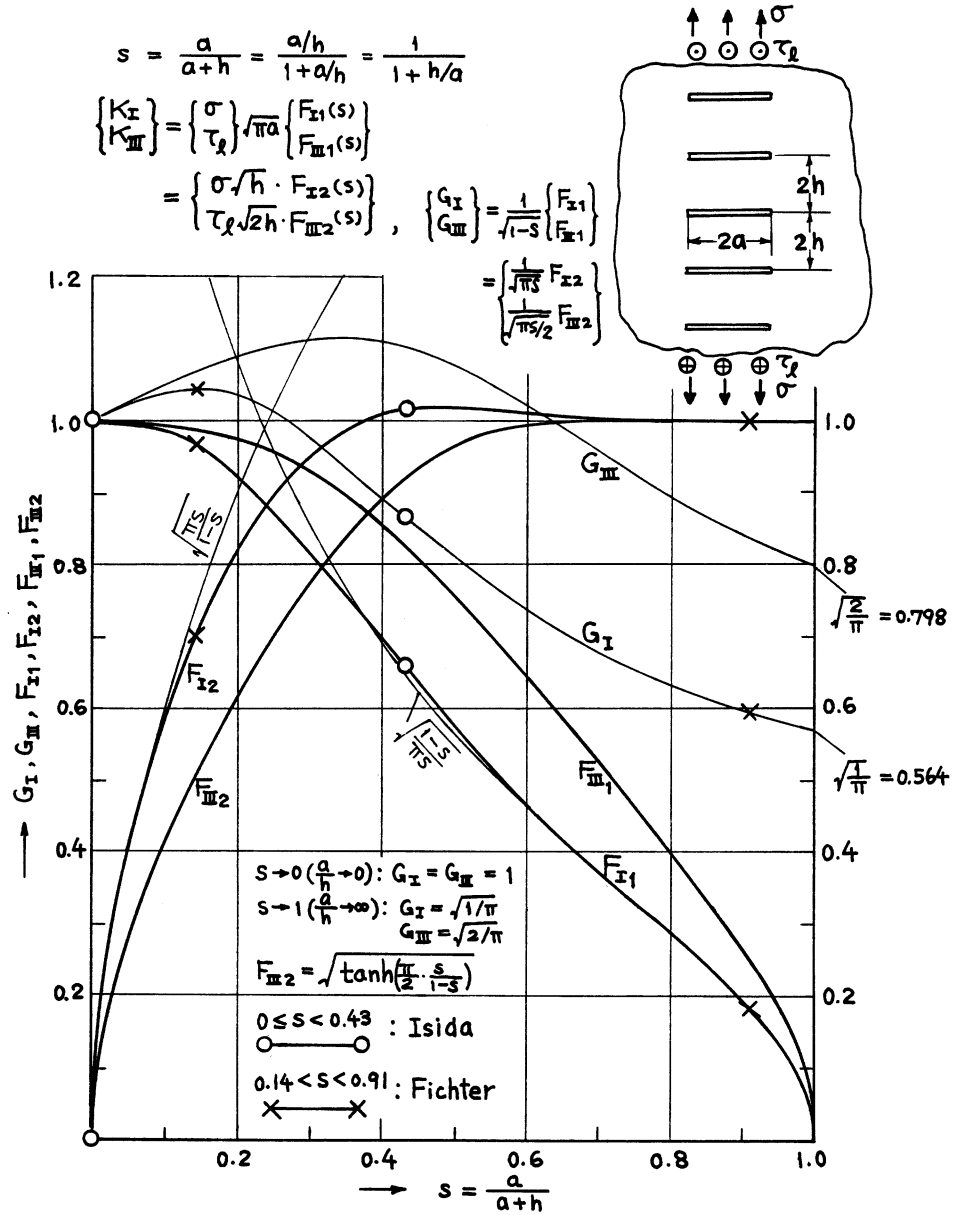
Reference: **Tada 1973**



$$\bar{Z}_{III}(z) = \frac{2t}{\pi} \int_a^b \tan^{-1} \sqrt{\frac{1 - \left(\sinh \frac{\pi a}{H} / \sinh \frac{\pi x_0}{H} \right)^2}{\left(\sinh \frac{\pi a}{H} / \sinh \frac{\pi z}{H} \right)^2 - 1}} dx_0$$

$$w(x, 0) = \frac{t}{\pi G} \int_a^b \ln \left| \frac{\sqrt{1 - \left(\sinh \frac{\pi a}{H} / \sinh \frac{\pi x_0}{H}\right)^2} + \sqrt{1 - \left(\sinh \frac{\pi a}{H} / \sinh \frac{\pi x}{H}\right)^2}}{\sqrt{1 - \left(\sinh \frac{\pi a}{H} / \sinh \frac{\pi x_0}{H}\right)^2} - \sqrt{1 - \left(\sinh \frac{\pi a}{H} / \sinh \frac{\pi x}{H}\right)^2}} \right| dx_0$$

Reference: Tada 1973



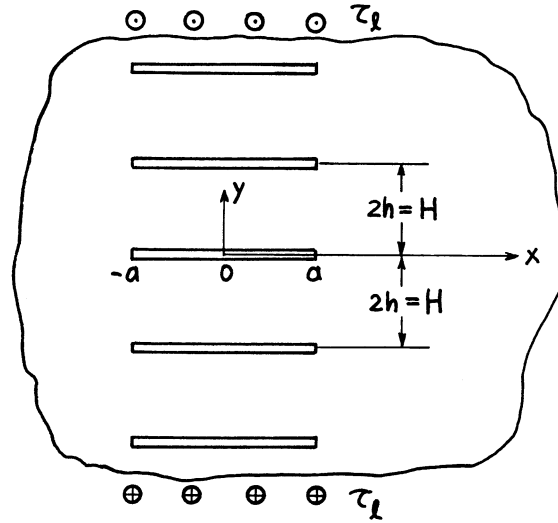
Method: K_I Fourier Transform (Fichter), Expansions of Complex Potentials (Isida), Asymptotic Approximation (polynomial of degree ten) (Benthem)

K_{III} Stress Concentration Factor (Neuber), Westergaard Stress Function (Tada)

Accuracy: K_I Estimated at 1%

K_{III} Exact

References: Fichter 1967; Isida 1971a; Benthem 1972; Neuber 1937; Tada 1973



$$z = x + iy$$

$$Z_{III}(z) = \frac{\tau_\ell}{\sqrt{1 - \left(\sinh \frac{\pi a}{H} / \sinh \frac{\pi z}{H} \right)^2}}$$

$$\bar{Z}_{III}(z) = \frac{\tau_\ell H}{\pi} \cosh^{-1} \left(\frac{\cosh \frac{\pi z}{H}}{\cosh \frac{\pi a}{H}} \right)$$

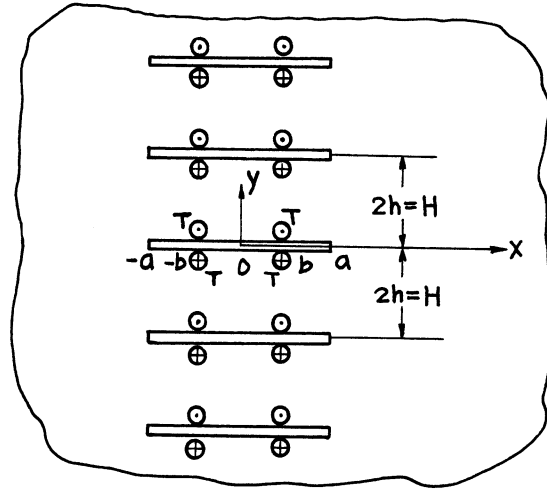
$$K_{III} = \tau_\ell \sqrt{H \tanh \frac{\pi a}{H}}$$

$$w(x, 0) = \frac{\tau_\ell}{G} \cdot \frac{H}{\pi} \cos^{-1} \left(\frac{\cosh \frac{\pi x}{H}}{\cosh \frac{\pi a}{H}} \right) \quad |x| \leq a$$

Method: Westergaard Stress Function

Accuracy: Exact

Reference: **Tada 1973**



$$z = x + iy$$

$$Z_{III}(z) = \frac{2T}{H} \cdot \frac{\cosh \frac{\pi b}{H} \sqrt{\left(\sinh \frac{\pi a}{H}\right)^2 - \left(\sinh \frac{\pi b}{H}\right)^2}}{\left\{ \left(\sinh \frac{\pi z}{H}\right)^2 - \left(\sinh \frac{\pi b}{H}\right)^2 \right\} \sqrt{1 - \left(\sinh \frac{\pi a}{H} / \sinh \frac{\pi z}{H}\right)^2}}$$

$$\bar{Z}_{III}(z) = \frac{2T}{\pi} \tan^{-1} \sqrt{\frac{1 - \left(\cosh \frac{\pi a}{H} / \cosh \frac{\pi b}{H}\right)^2}{\left(\cosh \frac{\pi a}{H} / \cosh \frac{\pi z}{H}\right)^2 - 1}}$$

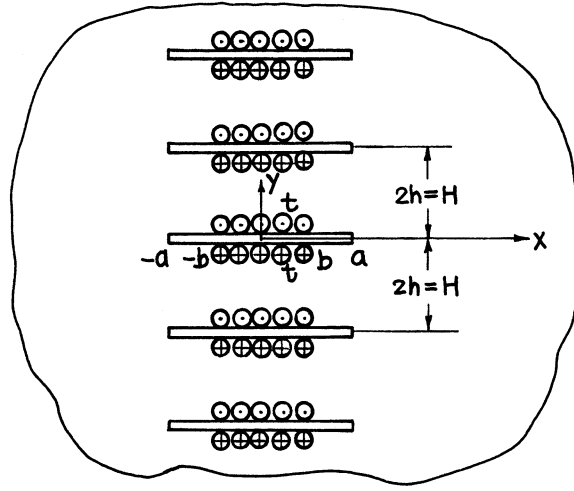
$$K_{III} = \frac{2T}{H} \sqrt{H \tanh \frac{\pi a}{H}} \frac{\cosh \frac{\pi b}{H}}{\sqrt{\left(\sinh \frac{\pi a}{H}\right)^2 - \left(\sinh \frac{\pi b}{H}\right)^2}}$$

$$w(x, 0) = \frac{2T}{\pi G} \left(\frac{\tanh^{-1}}{\coth^{-1}} \right) \sqrt{\frac{\left(\cosh \frac{\pi a}{H} / \cosh \frac{\pi b}{H}\right)^2 - 1}{\left(\cosh \frac{\pi a}{H} / \cosh \frac{\pi x}{H}\right)^2 - 1}} \quad \left(\begin{array}{l} |x| < b \\ b < |x| \leq a \end{array} \right)$$

Method: Westergaard Stress Function

Accuracy: Exact

Reference: **Tada 1973**



$$z = x + iy$$

$$Z_{III}(z) = \frac{2t}{\pi} \left\{ \frac{\sin^{-1} \left(\sinh \frac{\pi b}{H} / \sinh \frac{\pi a}{H} \right)}{\sqrt{1 - \left(\sinh \frac{\pi a}{H} / \sinh \frac{\pi z}{H} \right)^2}} + \tan^{-1} \frac{\left(\sinh \frac{\pi a}{H} / \sinh \frac{\pi b}{H} \right)^2 - 1}{\sqrt{1 - \left(\sinh \frac{\pi a}{H} / \sinh \frac{\pi z}{H} \right)^2}} \right\}$$

$$\bar{Z}_{III}(z) = \frac{2t}{\pi} \int_0^b \tan^{-1} \frac{1 - \left(\cosh \frac{\pi a}{H} / \cosh \frac{\pi x_0}{H} \right)^2}{\left(\cosh \frac{\pi a}{H} / \cosh \frac{\pi z}{H} \right)^2 - 1} dx_0$$

$$K_{III} = \frac{2t}{\pi} \sqrt{H \tanh \frac{\pi a}{H}} \sin^{-1} \left(\frac{\sinh \frac{\pi b}{H}}{\sinh \frac{\pi a}{H}} \right)$$

$$w(x, 0) = \frac{t}{\pi G} \int_0^b \ln \left| \frac{\sqrt{\left(\cosh \frac{\pi a}{H} / \cosh \frac{\pi x_0}{H} \right)^2 - 1} + \sqrt{\left(\cosh \frac{\pi a}{H} / \cosh \frac{\pi x}{H} \right)^2 - 1}}{\sqrt{\left(\cosh \frac{\pi a}{H} / \cosh \frac{\pi x_0}{H} \right)^2 - 1} - \sqrt{\left(\cosh \frac{\pi a}{H} / \cosh \frac{\pi x}{H} \right)^2 - 1}} \right| dx_0$$

Method: Westergaard Stress Function

Accuracy: Exact

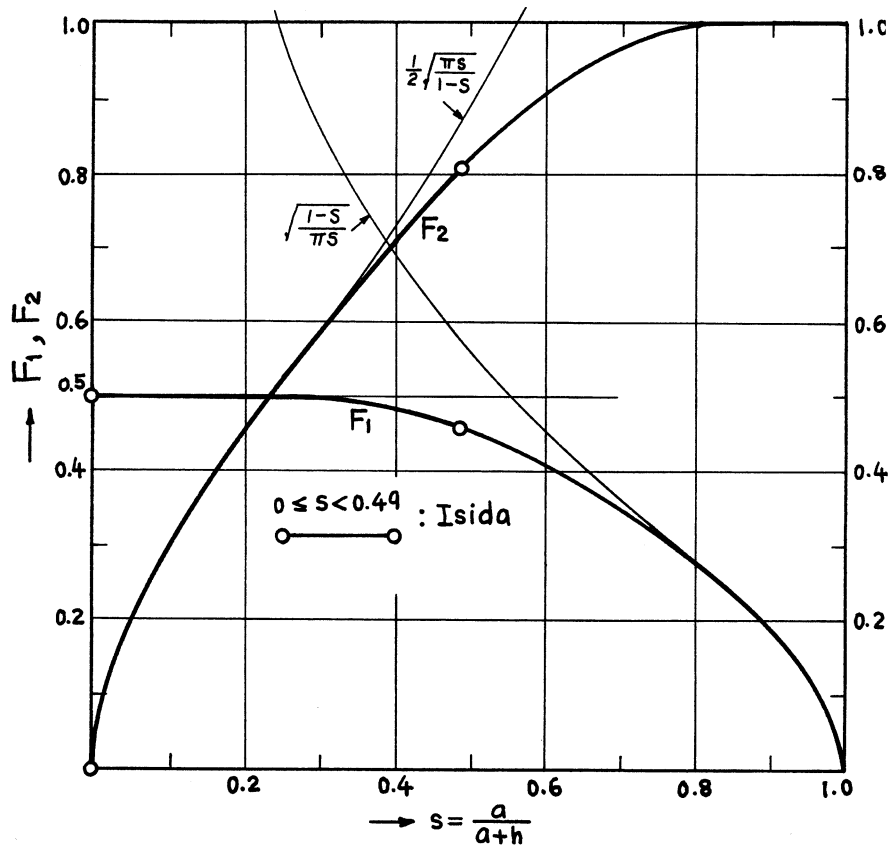
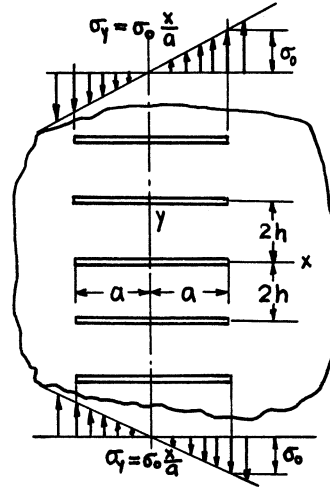
Reference: **Tada 1973**

$$s = \frac{a}{a+h} = \frac{a/h}{1+a/h} = \frac{1}{1+h/a}$$

$$\begin{aligned} K_I &= \sigma_0 \sqrt{\pi a} \cdot F_1(s) \\ &= \sigma_0 \sqrt{h} \cdot F_2(s) \end{aligned}$$

$$\begin{aligned} s \rightarrow 0 \left(\frac{a}{h} \rightarrow 0 \right) : F_1 &= \frac{1}{2} \\ F_2 &\rightarrow \frac{1}{2} \sqrt{\frac{\pi a}{h}} \end{aligned}$$

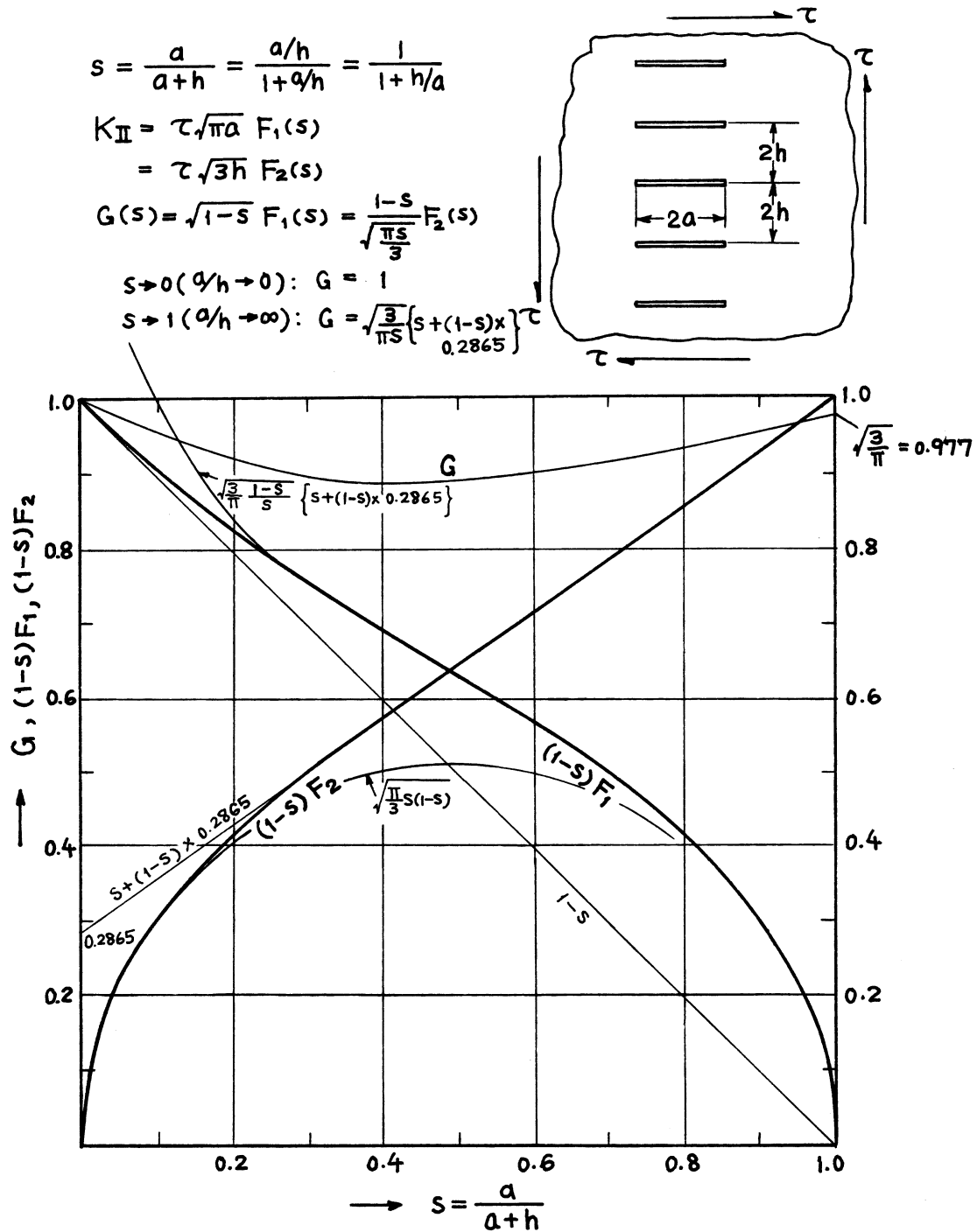
$$\begin{aligned} s \rightarrow 1 \left(\frac{a}{h} \rightarrow \infty \right) : F_2 &= 1 \\ F_1 &\rightarrow \sqrt{\frac{h}{\pi a}} \end{aligned}$$



Methods: Expansions of Complex Potentials ($s \leq 0.49$, Isida), Interpolated Asymptotically ($s \geq 0.49$, Tada)

Accuracy: Better than 1% for $s \leq 0.49$, estimated at 3% for $s \geq 0.49$.

References: **Isida 1971a; Tada 1973**



Methods: Asymptotic Approximation (Benthem), Dislocation Distribution (Kamei)

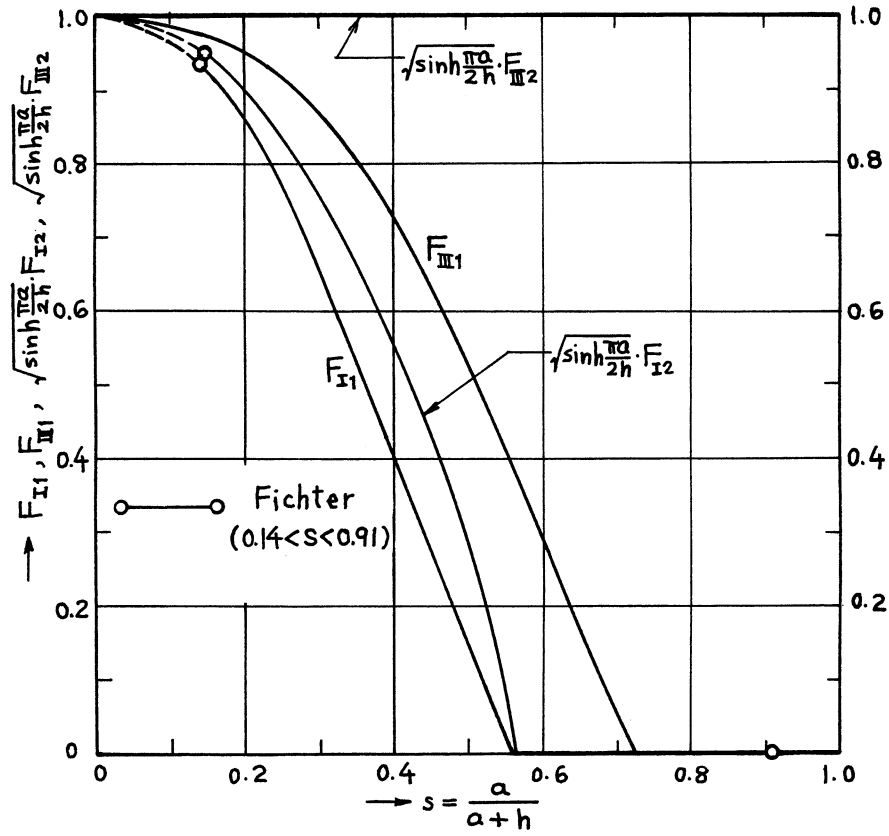
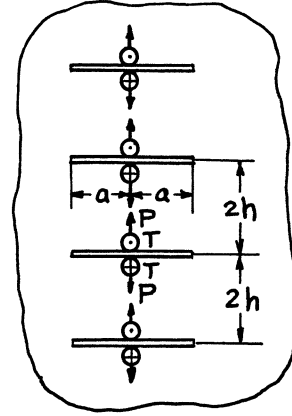
Accuracy: Better than 1%

References: Benthem 1972; Kamei 1974.

$$s = \frac{a}{a+h} = \frac{a/h}{1+a/h} = \frac{1}{1+h/a}$$

$$\begin{aligned} \begin{Bmatrix} K_I \\ K_{III} \end{Bmatrix} &= \frac{1}{\sqrt{\pi a}} \begin{Bmatrix} P \\ T \end{Bmatrix} \begin{Bmatrix} F_{I1}(s) \\ F_{III1}(s) \end{Bmatrix} \\ &= \frac{1}{\sqrt{2h}} \begin{Bmatrix} P \\ T \end{Bmatrix} \begin{Bmatrix} F_{I2}(s) \\ F_{III2}(s) \end{Bmatrix} \end{aligned}$$

$$F_{III2} \equiv 1 / \sqrt{\sinh \frac{\pi a}{2h}}$$



Methods: K_I Fourier Transform (Fichter)

K_{III} Westergaard Stress Function (Tada)

Accuracy: K_I 1% ,

K_{III} Exact

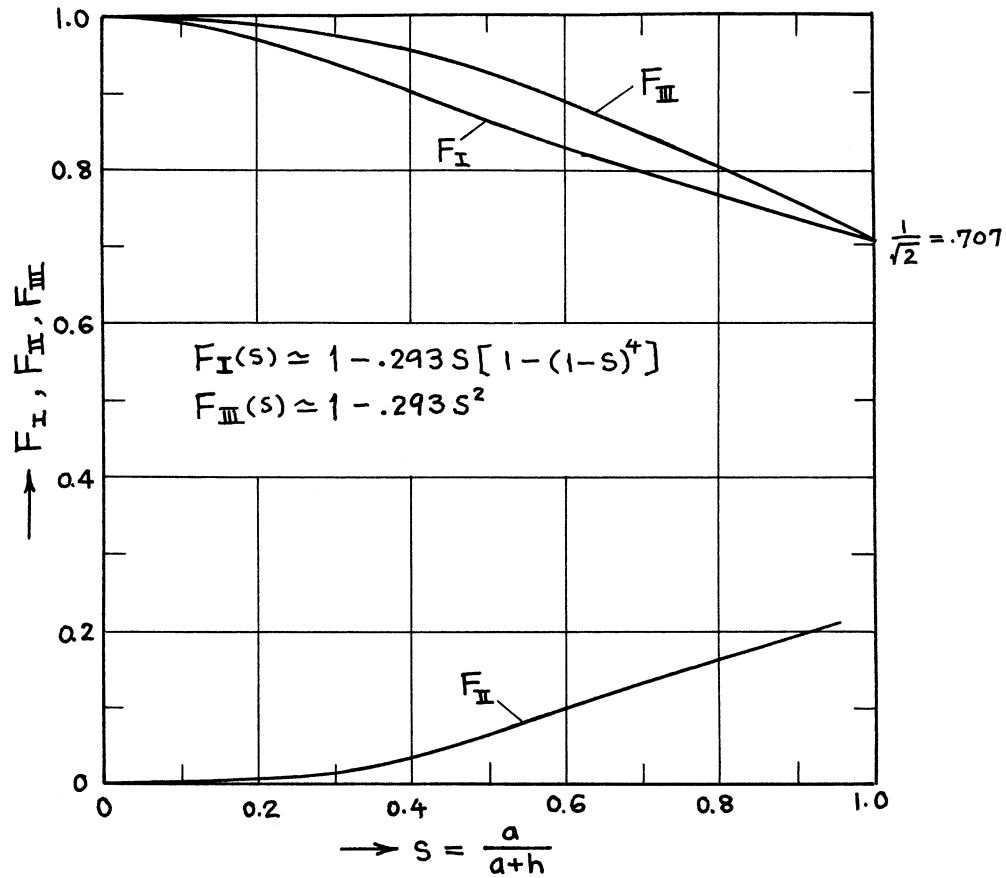
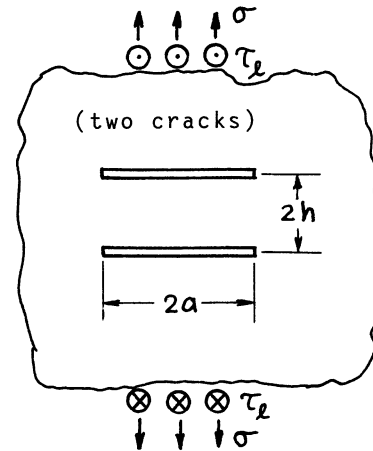
References: **Fichter 1967; Tada 1973**

$$K_I = \sigma \sqrt{\pi a} \cdot F_I(s)$$

$$K_{II} = \sigma \sqrt{\pi a} \cdot F_{II}(s)$$

$$K_{III} = \tau_\ell \sqrt{\pi a} \cdot F_{III}(s)$$

$$s = \frac{a}{a+h}$$

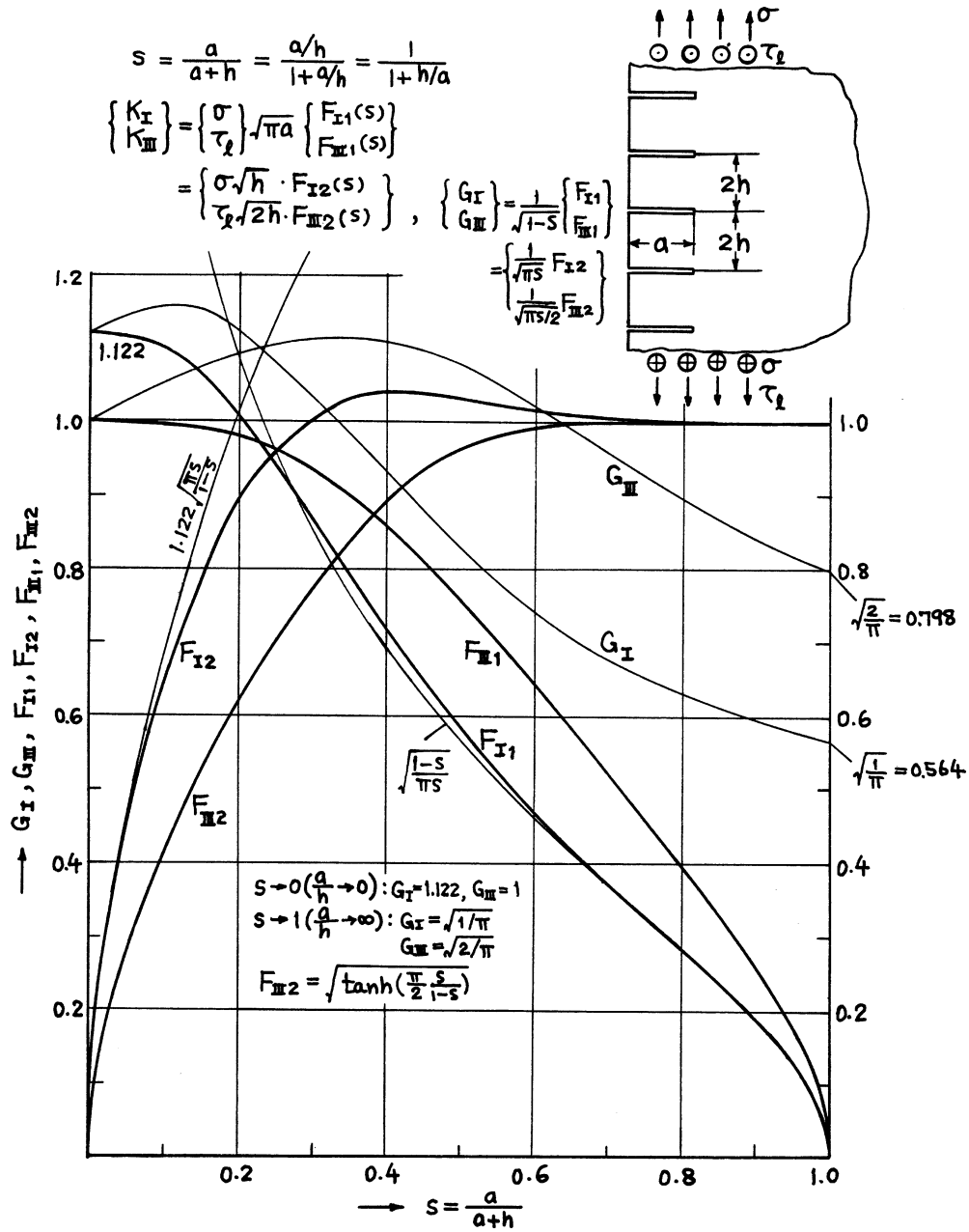


Methods: Series Expansion of Complex Potentials (Isida, $0 < s < 1/2$), Dislocation Distribution (Kamei)

Accuracy: F_I and F_{III} Better than 1%

F_{II} Expected to be better than 5%

References: **Isida 1972; Kamei 1974; Tada 1985**

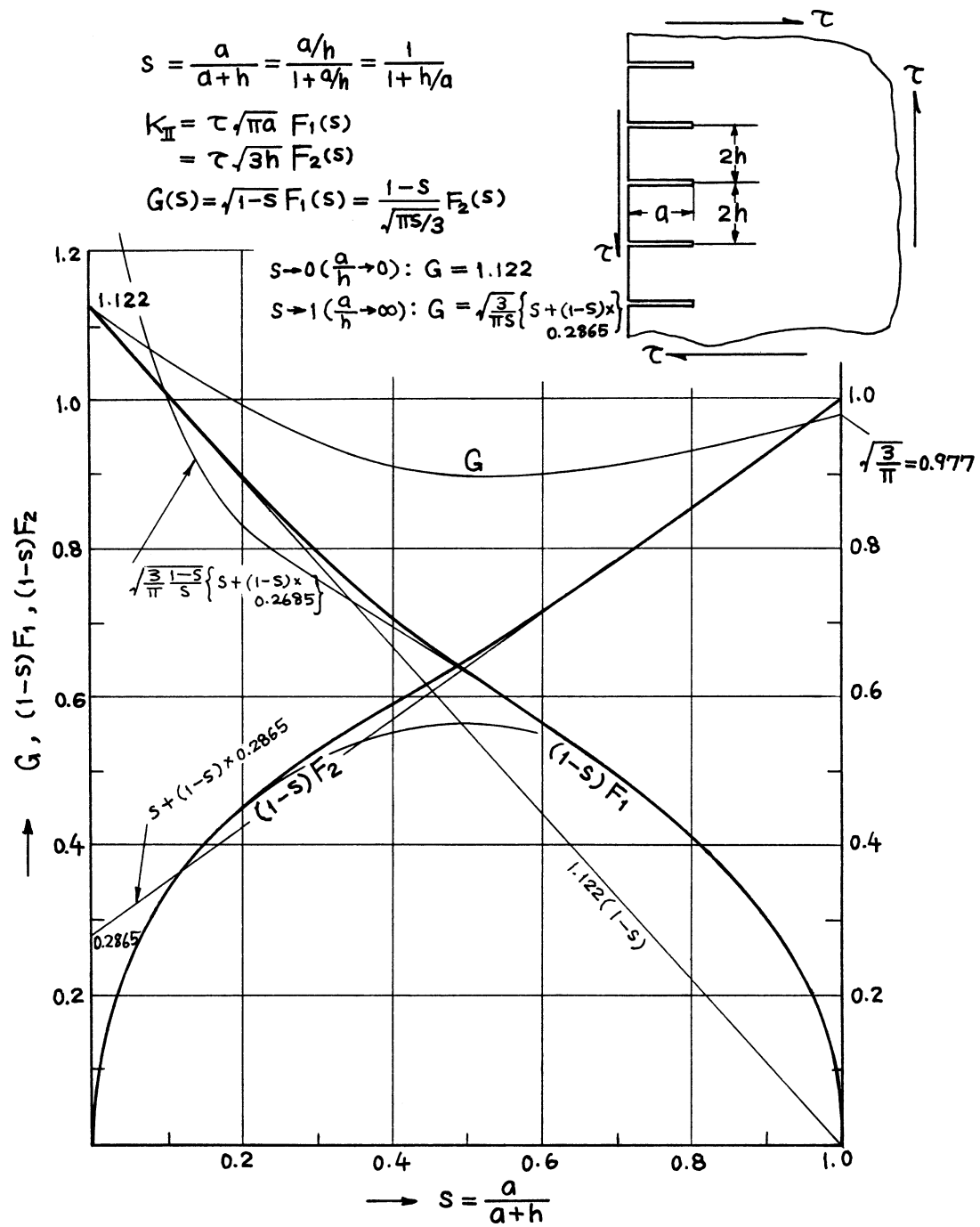


Methods: K_I Asymptotic Interpolation (Benthem), Mapping Function Method (Bowie),
 Body Force Method (Nishitani); K_{III} Stress Concentration Factor (Neuber),
 Westergaard Stress Function (Tada)

Accuracy: K_I 1% ,

K_{III} Exact

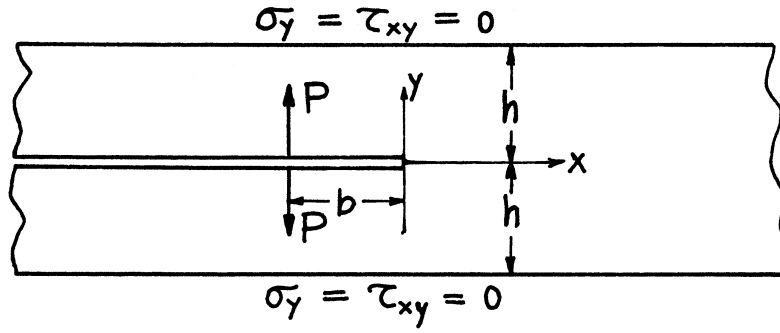
References: Neuber 1937, 1958; Benthem 1972; Tada 1973; Bowie 1973; Nishitani 1976



Method: Estimated Asymptotically

Accuracy: Estimated at better than 3%

Reference: **Tada 1973**



$$s = \frac{b}{b+h} = \frac{b/h}{1+b/h} = \frac{1}{1+h/b}$$

$$\begin{aligned} K_I &= \sqrt{\frac{2}{\pi}} \frac{P}{\sqrt{b}} \cdot F_1(s) \\ &= \frac{P}{\sqrt{h}} \left(\frac{b}{h} \right) F_2(s) \end{aligned}$$

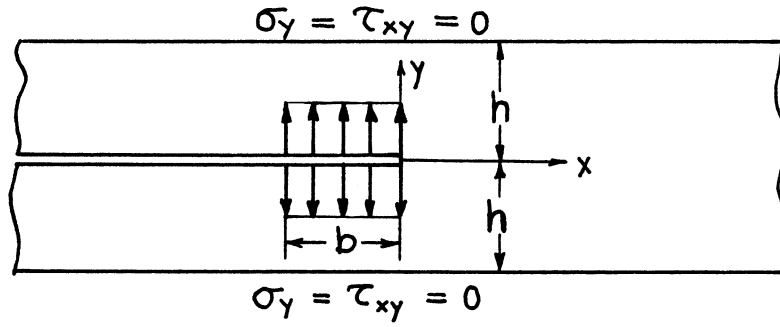
$$\begin{aligned} s \rightarrow 0 \left(\frac{b}{h} \rightarrow 0 \right) : F_1 &= 1 \\ F_2 &\rightarrow \sqrt{\frac{2}{\pi}} \left(\frac{h}{b} \right)^{3/2} = \sqrt{\frac{2}{\pi}} \left(\frac{1-s}{s} \right)^{3/2} \end{aligned}$$

$$\begin{aligned} s \rightarrow 1 \left(\frac{b}{h} \rightarrow \infty \right) : F_2 &= 2\sqrt{3} \\ F_1 &\rightarrow \sqrt{6\pi} \left(\frac{b}{h} \right)^{3/2} = \sqrt{6\pi} \left(\frac{s}{1-s} \right)^{3/2} \end{aligned}$$

Methods: $s \rightarrow 0$ Solution for a Semi-Infinite Crack (**page 3.6**), $s \rightarrow 1$ Theory of Beam Bending

Accuracy: Exact for both $s \rightarrow 0$ and $s \rightarrow 1$

References: $s \rightarrow 0$ **Irwin 1957, etc.**, $s \rightarrow 1$ **Gilman 1959; Barenblatt 1962**



$$s = \frac{b}{b+h} = \frac{b/h}{1+b/h} = \frac{1}{1+h/b}$$

$$\begin{aligned} K_I &= 2\sqrt{\frac{2}{\pi}} \sigma \sqrt{b} \cdot F_1(s) \\ &= \sigma \sqrt{h} \left(\frac{b}{h}\right)^2 \cdot F_2(s) \end{aligned}$$

$$s \rightarrow 0 \left(\frac{b}{h} \rightarrow 0\right) : F_1 = 1$$

$$F_2 \rightarrow 2\sqrt{\frac{2}{\pi}} \left(\frac{b}{h}\right)^{3/2} = 2\sqrt{\frac{2}{\pi}} \left(\frac{1-s}{s}\right)^{3/2}$$

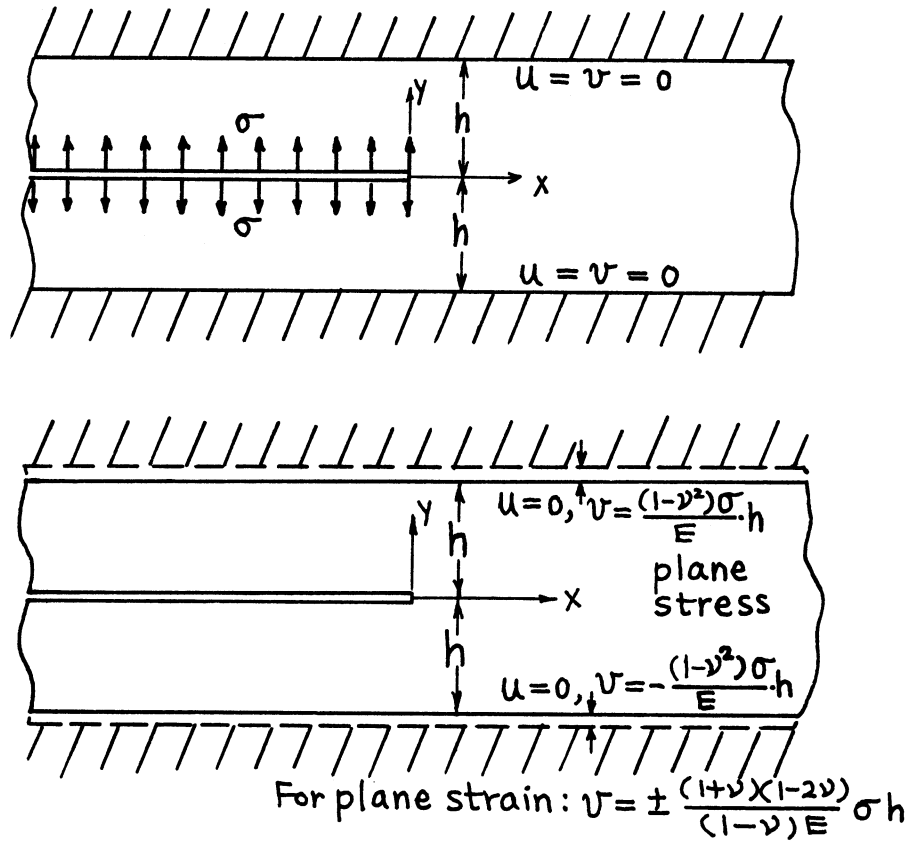
$$s \rightarrow 1 \left(\frac{b}{h} \rightarrow \infty\right) : F_2 = \sqrt{\frac{3}{2}}$$

$$F_1 \rightarrow \frac{\sqrt{3\pi}}{4} \left(\frac{b}{h}\right)^{3/2} = \frac{\sqrt{3\pi}}{4} \left(\frac{s}{1-s}\right)^{3/2}$$

Methods: $s \rightarrow 0$ Solution for a Semi-Infinite Crack (**page 3.7**), $s \rightarrow 1$ Theory of Beam Bending

Accuracy: Exact for both $s \rightarrow 0$ and $s \rightarrow 1$

References: $s \rightarrow 0$ **Tada 1973**, $s \rightarrow 1$ **Gilman 1959; Barenblatt 1962**



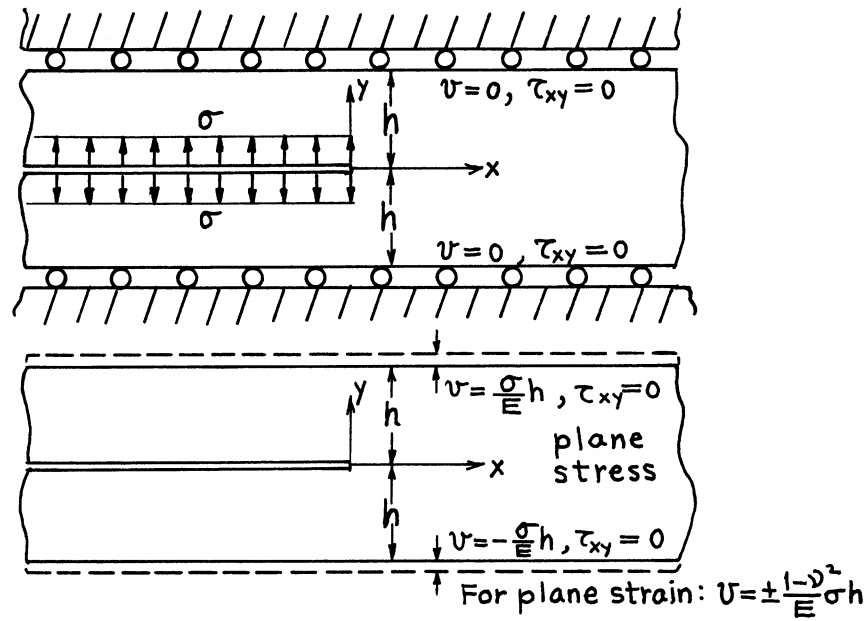
$$K_I = \sqrt{1 - \nu^2} \cdot \sigma \sqrt{h} \quad \text{plane stress}$$

$$K_I = \frac{\sqrt{1 - 2\nu}}{1 - \nu} \cdot \sigma \sqrt{h} \quad \text{plane strain}$$

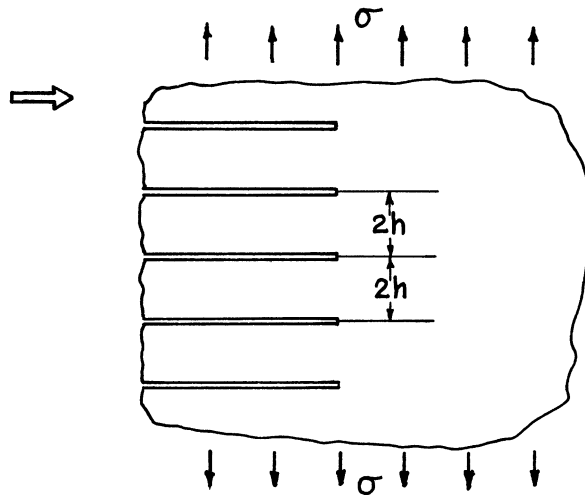
Method: Energy Consideration

Accuracy: Exact

Reference: **Rice 1967**



$$K_I = \sigma \sqrt{h}$$

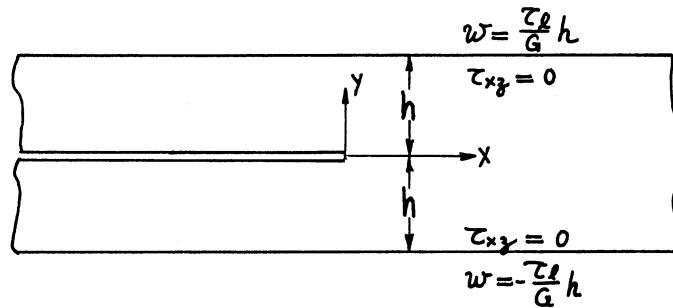
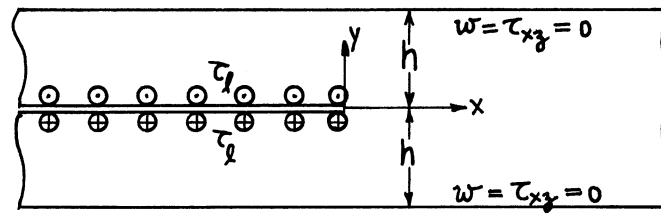


Methods: Energy Consideration (Paris, Rice), Fourier Transform (Benthem)

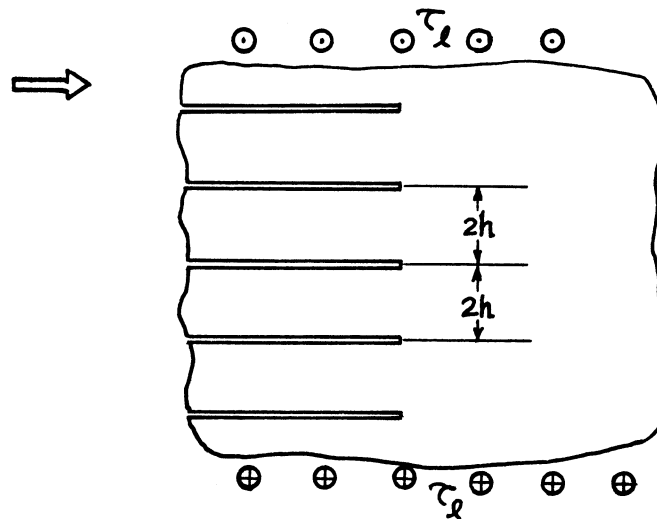
Accuracy: Exact

References: Paris 1955, 1960; Rice 1967; Benthem 1972

See page 12.1.



$$K_{III} = \tau_\ell \sqrt{2h}$$

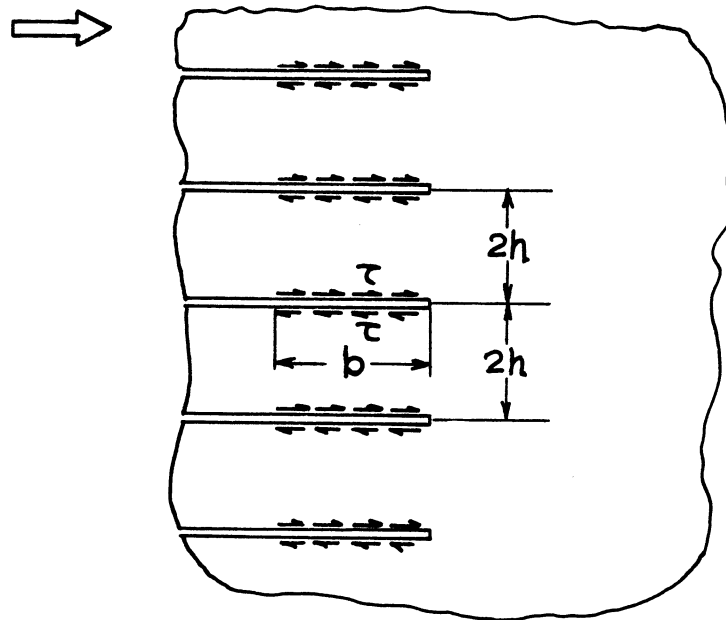
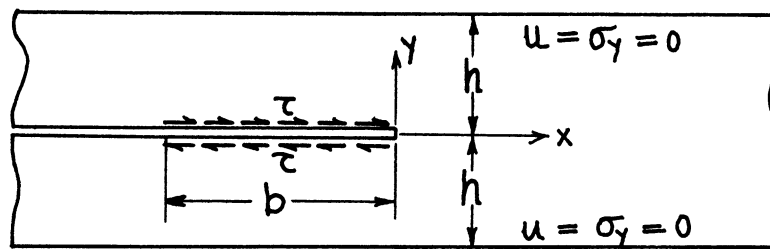


Method: Westergaard Stress Function

Accuracy: Exact

Reference: **Tada 1973**

See **page 12.2**.



See pages 12.5 and 12.6.

$$s = \frac{a}{a+h} = \frac{a/h}{1+a/h} = \frac{1}{1+h/a}$$

$$K_I = \sigma \sqrt{\pi a} \cdot F_1(s)$$

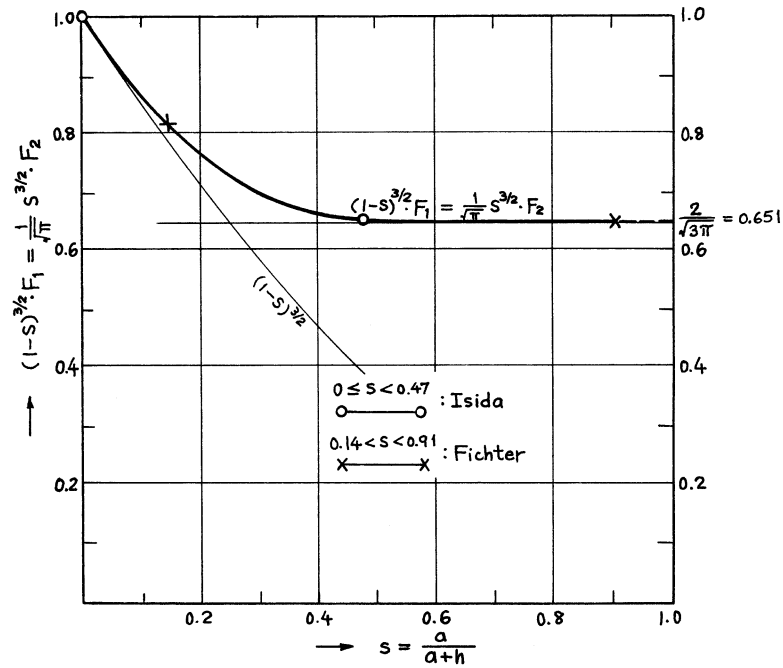
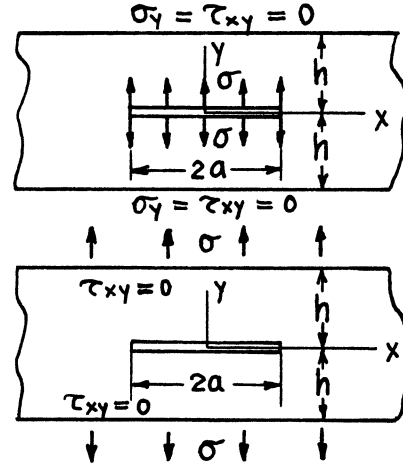
$$= \sigma \sqrt{h} \left(\frac{a}{h}\right)^2 \cdot F_2(s)$$

$$s \rightarrow 0 (a/h \rightarrow 0) : F_1 = 1$$

$$F_2 \rightarrow \sqrt{\pi} \left(\frac{h}{a}\right)^{3/2}$$

$$s \rightarrow 1 (a/h \rightarrow \infty) : F_2 = 2/\sqrt{3}$$

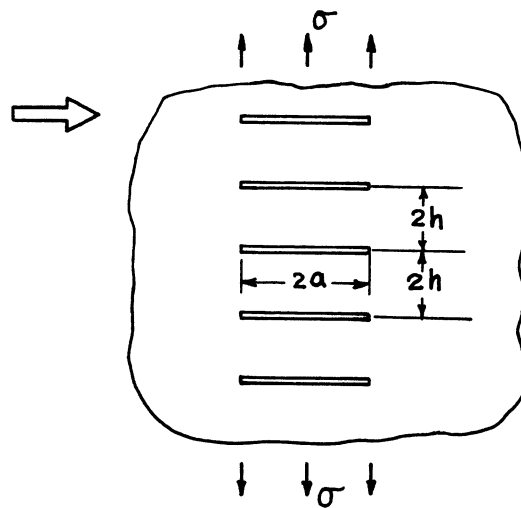
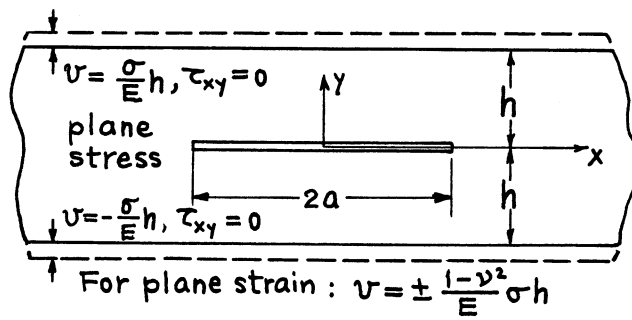
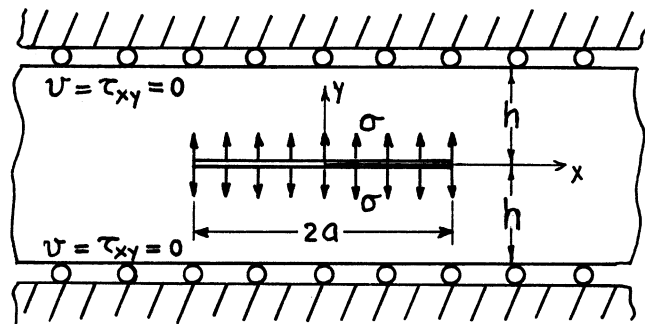
$$F_1 \rightarrow \frac{2}{\sqrt{3\pi}} \left(\frac{a}{h}\right)^{3/2}$$



Methods: Expansions of Complex Stress Potentials (Isida), Fourier Transform (Fichter), Theory of Beam Bending ($s \rightarrow 1$)

Accuracy: Order of 1%

References: **Fichter 1967; Isida 1971a**



See page 14.1.

$$s = \frac{a}{a+h} = \frac{a/h}{1+a/h} = \frac{1}{1+h/a}$$

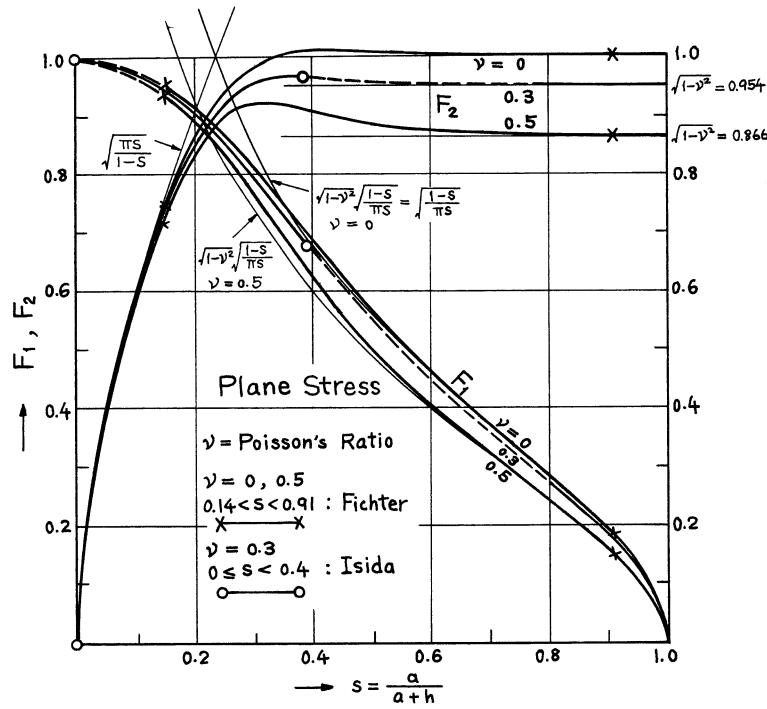
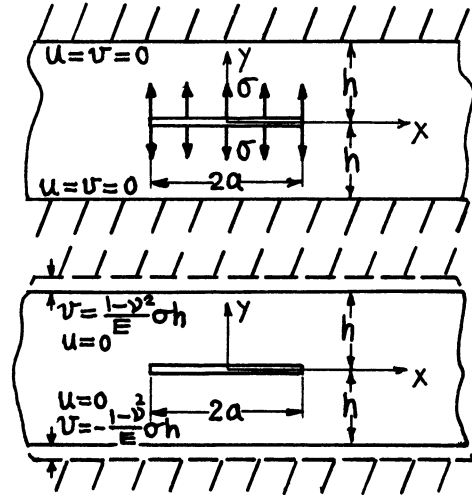
$$K_I = \sigma\sqrt{\pi a} \cdot F_1(s)$$

$$= \sigma\sqrt{h} \cdot F_2(s)$$

$$s \rightarrow 0 (a/h \rightarrow 0) : F_1 = 1, F_2 \rightarrow \sqrt{\frac{\pi a}{h}}$$

$$s \rightarrow 1 \left(\frac{a}{h} \rightarrow \infty \right) : F_2 = \sqrt{1-\nu^2}$$

$$F_1 \rightarrow \sqrt{1-\nu^2} \sqrt{\frac{h}{\pi a}}$$



Methods: Expansions of Complex Stress Potentials (Isida), Fourier Transform (Fichter), $s \rightarrow 0$: Solution for Infinite Plate, $s \rightarrow 1$: Energy Balance (Rice)

Accuracy: Order of 1%

References: **Fichter 1967; Rice 1967; Isida 1971a**

NOTE: For plane strain $F_2(s \rightarrow 1) = \sqrt{1-2\nu/(1-\nu)}$, etc.

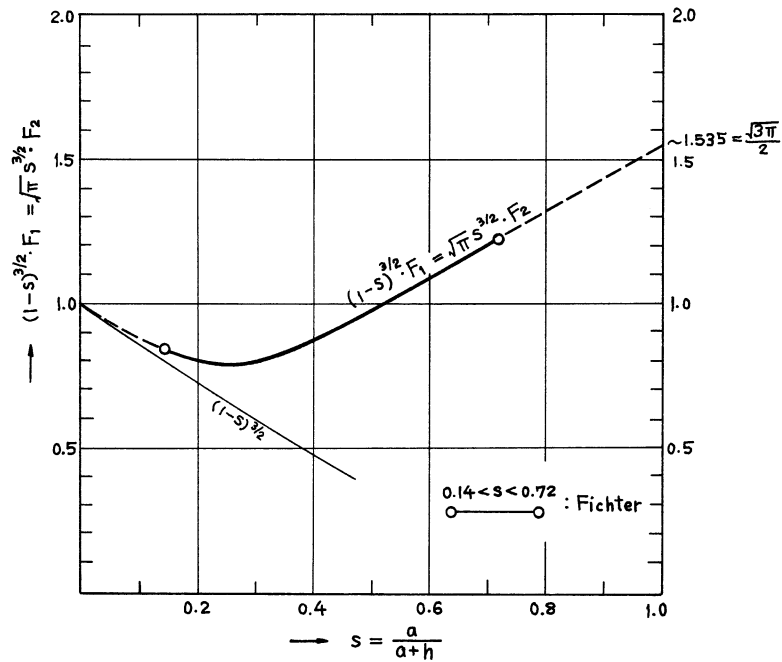
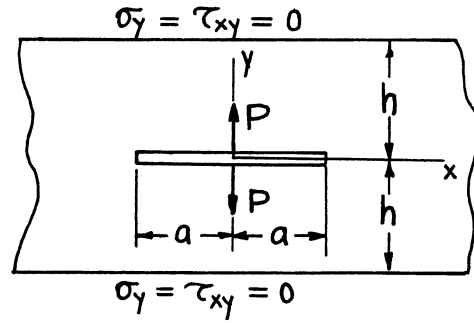
$$s = \frac{a}{a+h} = \frac{a/h}{1+a/h} = \frac{1}{1+h/a}$$

$$\begin{aligned} K_I &= \frac{P}{\sqrt{\pi a}} \cdot F_1(s) \\ &= \frac{P}{\sqrt{h}} \left(\frac{a}{h} \right) \cdot F_2(s) \end{aligned}$$

$$s \rightarrow 0 \left(\frac{a}{h} \rightarrow 0 \right) : F_1 = 1, F_2 \rightarrow \frac{1}{\sqrt{\pi}} \left(\frac{h}{a} \right)^{3/2}$$

$$s \rightarrow 1 \left(\frac{a}{h} \rightarrow \infty \right) : F_2 = \frac{\sqrt{3}}{2}$$

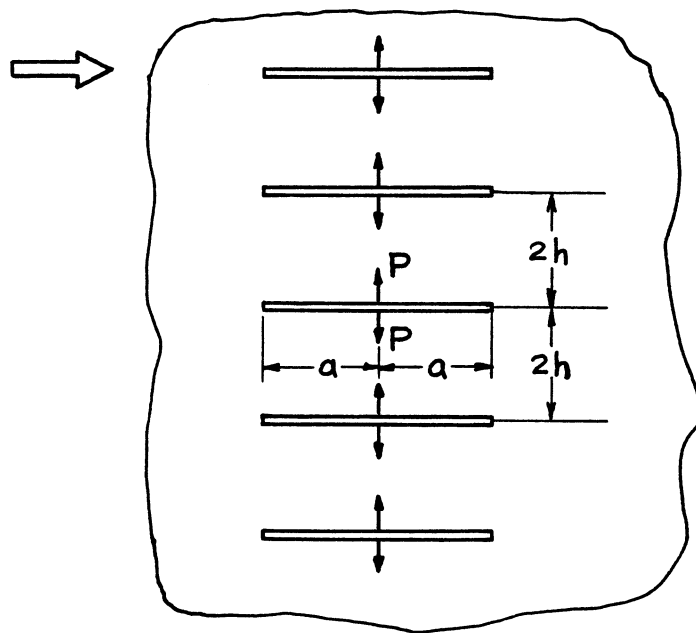
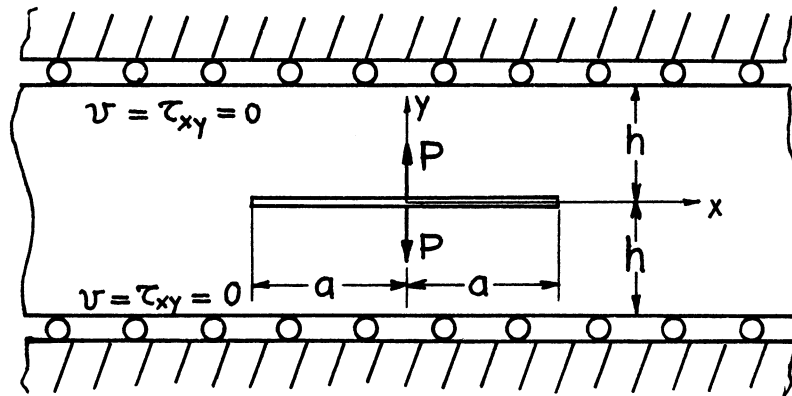
$$F_1 \rightarrow \frac{\sqrt{3\pi}}{2} \left(\frac{a}{h} \right)^{3/2}$$



Methods: Fourier Transform (Fichter), $s \rightarrow 0$: Solution for infinite plate (**page 5.9**), $s \rightarrow 1$: Beam bending

Accuracy: Order of 1%

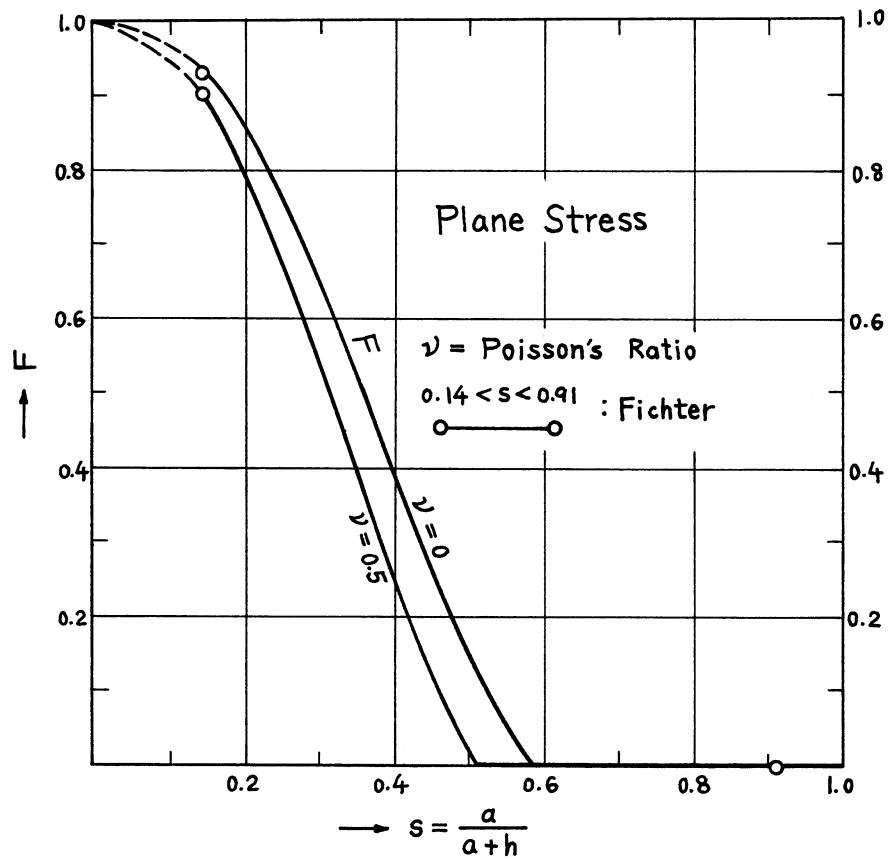
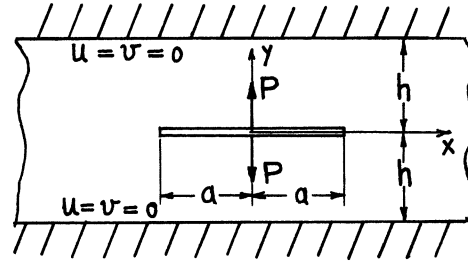
Reference: **Fichter 1967**



See page 14.6.

$$s = \frac{a}{a+h} = \frac{a/h}{1+a/h} = \frac{1}{1+h/a}$$

$$K_I = \frac{P}{\sqrt{\pi a}} \cdot F(s)$$



Methods: Fourier Transform (Fichter), $s \rightarrow 0$: Solution for an Infinite Plate (page 5.9)

Accuracy: Order of 1%

Reference: **Fichter 1967**

$$s = \frac{a}{a+h} = \frac{a/h}{1+a/h} = \frac{1}{1+h/a}$$

$$g\left(\frac{b}{a}\right) = \sqrt{1 - \left(\frac{b}{a}\right)^2}$$

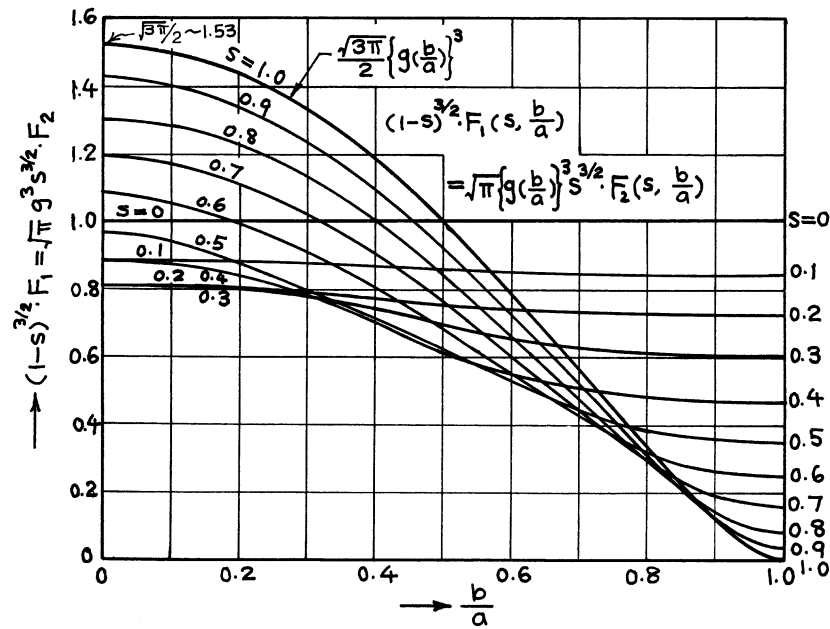
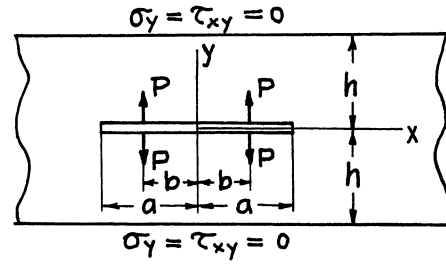
$$\begin{aligned} K_I &= \frac{2P}{\sqrt{\pi a}} \cdot \frac{1}{g(b/a)} \cdot F_1\left(s, \frac{b}{a}\right) \\ &= \frac{2P}{\sqrt{h}} \left(\frac{a}{h}\right) \left\{g\left(\frac{b}{a}\right)\right\}^2 F_2\left(s, \frac{b}{a}\right) \end{aligned}$$

$$s \rightarrow 0 \left(\frac{a}{h} \rightarrow 0\right) : F_1 = 1$$

$$F_2 \rightarrow \frac{1}{\sqrt{\pi}} \cdot \frac{1}{g^3} \left(\frac{h}{a}\right)^{3/2}$$

$$s \rightarrow 1 \left(\frac{a}{h} \rightarrow \infty\right) : F_2 = \sqrt{3}/2$$

$$F_1 \rightarrow \frac{\sqrt{3\pi}}{2} \cdot g^3 \cdot \left(\frac{a}{h}\right)^{3/2}$$

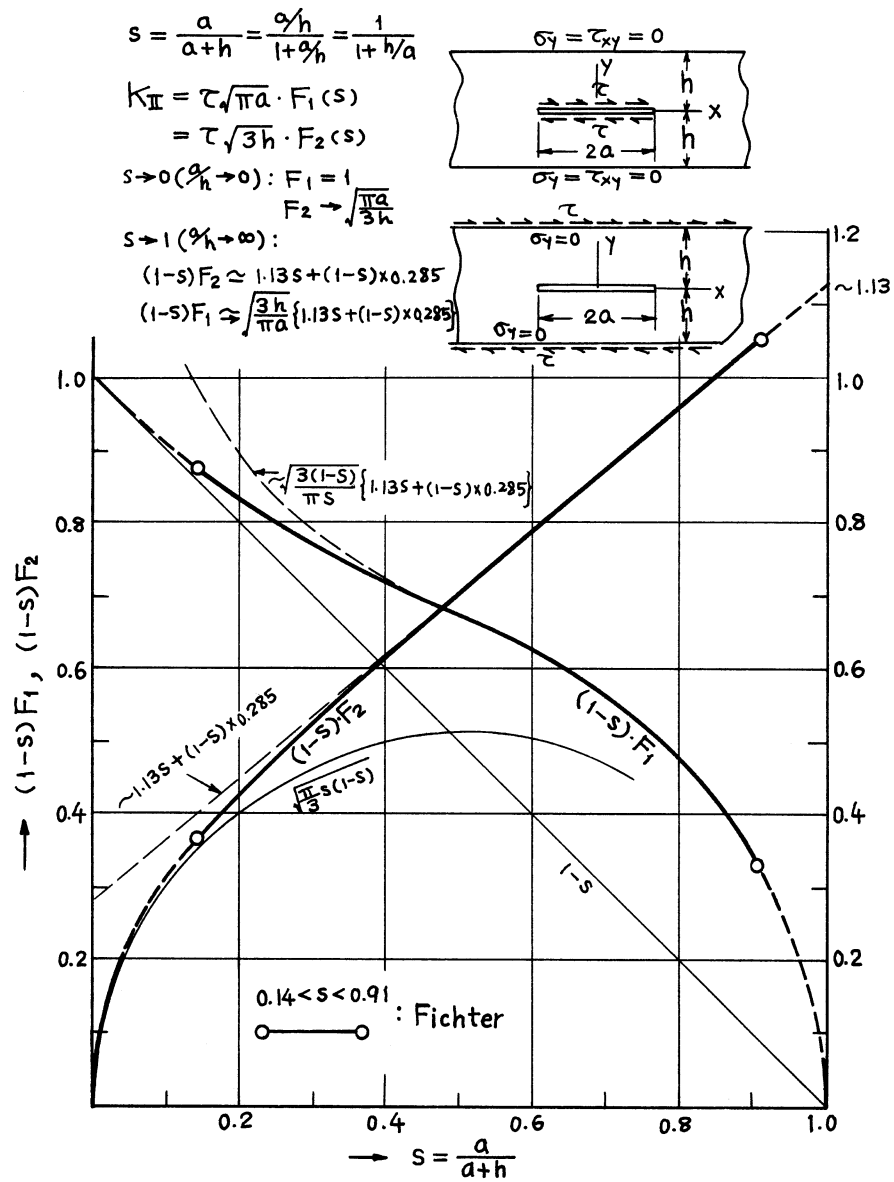


Method: Interpolated Asymptotically

Accuracy: Expected to be better than 5% for any s and b/a

Reference: **Tada 1973**

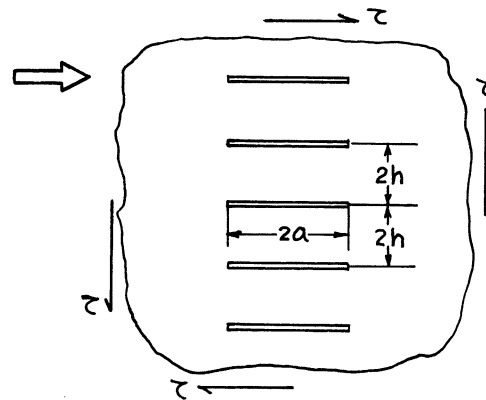
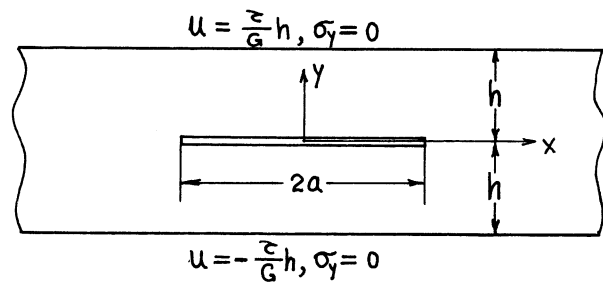
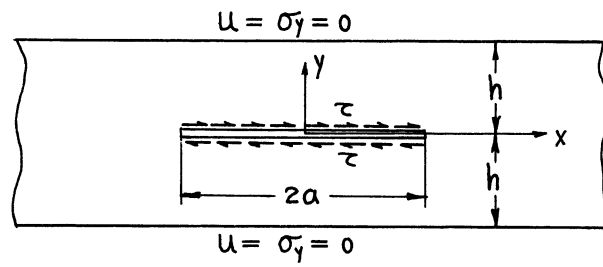
NOTE: $F(s \rightarrow 0, b/a)$ (page 5.11), $F(s \rightarrow 1, b/a)$ (beam bending), $F(s, 0)$ (page 17.4), $F(s, b/a \rightarrow 1)$ (page 3.6).



Method: Fourier Transform

Accuracy: Order of 1%

Reference: **Fichter 1967**



See page 14.5.